KALASALINGAM UNIVERSITY

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

ELECTROCOMM -6

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EDITOR'S MESSAGE

As being the Head of the Department and Editor of this Magazine "ELECTROCOMM", I am pleased to wish all my students who contributed their hard work and valuable time in designing this wonderful magazine. We keep on introducing new platforms for our students to develop them to be augmented in knowledge. In that front, we are releasing the sixth issue of our department magazine ELECTROCOMM. "Practice makes the man perfect" is a quote. Yup! We have practiced a lot and refined ourselves in the work done. I appreciate the successful and consistent efforts of the editors for releasing this magazine for the past three years.

Finally, I conclude my message by thanking the entire team and I wish all the students to make this effort as a successful one.

Dr. S. Durairaj,
Chief Editor.

MESSAGE

First of all, I wish to express the happiness. I feel to see the commitment of faculty members and students of the department of ECE to regularly bringforth the magazine.

When I was first approached by Mr.V. Vedhanarayanan, B.Tech (Third year) ECE with a request to write a message for the magazine, I wondered how from a Civil Engg., background, would be able to write a meaningful message. Mr.V.Vedhanarayanan shared with me briefly the information content of the magazine and how this volume is going to address or at least discuss some of the recent undesirable, unpredictable social events such as the brutal murder of school teacher Mrs. Uma by a 9th grade student. It's from here; I got an idea about what message to share with all of you.

Yes, with the advancement of technology, so many comforts are better. But sadly the most vital part of systematic development of character in a child or student is completely neglected. Often parents do emphasize on simple moral values or lessons but without a systematic education in schools/colleges very soon. Such home level training takes a back seat what the individual is challenged with circumstances of the life.

I used to wonder during my childhood what is the most fundamental part of character with which all other good qualities will manifest in an individual. I have seen people who manifest very good qualities during a particular circumstance and they behave completely opposite when the circumstances are changed. Why is that so?

The answer to this question, I got through one of my teachers who taught me what goes in forming deep character in the heart of an individual. The most fundamental quality to be cultivated is a deep sense of gratitude. My teacher says that "Feeling thank you!" which can be taught even to a parrot. One has to be trained to feel thank you at all any other good qualities has to manifest in that person.

We are seeing many cases of plagiarism being reported throughout the world. Of course, one can put many rules and regulations to stop such things, develop computer software that can quickly detect such an attempt etc. But ultimately speaking, it lives with an individual character of the person not to do such things. Suppose the student was to "feel" in his heart for the help extended by his supervisor, when a thought to cheat or manipulate comes in mind, it will prick his heart" How can I do this?"This will bring bad name for my supervisor!" This is possible if the student has trained himself to feel for the help received and not just made a habit of saying "thanks" only in words.

Take the case of Mrs. Uma. I am pretty sure she would have definitely given so many helps and affection to that boy. But 'new' has taught him how to be grateful for all that. We are given completely opposite education through cinemas and movie. If true sense of gratitude is taught right from childhood, the world we live in will be completely a different place.

I take the opportunity to request all my student friends to practice this owe principle deeply in life. It will bring lot of meaning in your exchanges in the teachers, parents, friends and in future, your spouse and children. This is the most important need of hour. I am personally trying to practice this principle in my life and although I fail so many times, I still try harder. This is bringing meaning to my life.

I sincerely extend my gratitude to the HOD/ECE, all the faculty members and the students to give this opportunity to share some thoughts.

Dr .C. Sivapragasam,
Prof & Head, Department of Civil Engg.,

MESSAGE

It gives me an immense pleasure to note that the Department of Electronics and Communication Engineering is releasing the 6th Issue of the department Magazine "ELECTROCOMM" during First Week of April 2012. Also, it is a matter of great delight that the students of both Under Graduate and Post Graduate Program are actively participating, sharing their novel ideas and thoughts in the Magazine. I hope this excellent initiative by the Department of Electronics and Communication Engineering in our University would create a good network among the students community and also would encourage them to pursue their higher education especially in their advanced domains of Electronics and Communication Engineering. I am sure that, the contents of this magazine will certainly stimulate all the students to achieve unprecedented landmarks in their specialization. I wholeheartedly appreciate all the sincere constructive efforts and meticulous planning of the entire team which has contributed its best to bring out the magazine "ELECTROCOMM" (Issue: 6) in virtue and in good shape.

Dr. S. Rajakarunakaran,
Prof & Head, Department of Mechanical Engg.,

MESSAGE

I want each and every student to read, read …. read. Once they develop a habit of reading, we the faculty feel a sense of achievement. I mention reading as a habit; a habit is like brushing our teeth every day. Knowing how to read and not reading is a curse, a sin.

Once we decide to read, the next question to arise is “What to read?”

I once listened to an audio lecture by Mr. Sukisivam in which he says,**“நீ கடையில் போய் புரட்டி புரட்டி வாங்கும் புத்தகம் சிறந்த புத்தகம் அல்ல. உன்னை எந்த புத்தகம் புரட்டி புரட்டி எடுக்கிறதோ அதுவே சிறந்த புத்தகம்”**. In English, “What you search and find in a bookshop is not a good book. But a book which makes a complete transformation in you is the best book”.
I want everyone to start his own personal library at home/hostel/workplace.

In our country, we give less importance to time management, and it is unfortunate that we need to wait in many places. I think the waiting is a bliss or gift to us. In an optimistic thinking, I feel in the places of waiting, we can read more.

Wherever you go, carry some books with you to read. You read whatever you like, let it be technical/non-technical/self-help/literature etc., Whatever language you like, be it your mother tongue or others, read good books. Nowadays, you can think of latest items like Amazon e-reader to enhance your reading habit.

In tamil there is a saying, “**கண்டது கற்க பண்டிதனாவான்**”which means “By learning too many things, one can become an expert”.

Whenever you get a chance to give gifts, always present best books (with your wisdom) to your friends / relatives / guests. I hope almost everyone will accept a book as most memorable gift.

Once you read, your mind and knowledge expand and it is not possible to shrink them back. Definitely there will be an expansion. Keep some of your pocket money for investing on good books.

A popular joke is there about the library that a group of people decided to go to a place for a picnic and they decided to go to the library which is the place normally nobody goes.

If you read a minimum of 10 pages a day, in 365 days, approximately you can read 3600 pages, may be 12 books in a year. In four years of your UG course, you must have a list of at least 50 books to be read other than your subjects before you leave the campus.

It is not what we read that makes us knowledgeable, but what we understand that makes us knowledgeable.

A teacher succeeds, if he is defeated by his own student.

Today a Reader, Tomorrow a Leader (W Fusselman)

Dr. S. Kannan,

Prof & Head, Department of Electrical & Electronics Engg.,

INSPIRATION

Mr.T.SENTHIL
Assistant Professor, ECE Department,
Kalasalingam University.

EDUCATION:

B.E(ECE)	:Madurai Kamaraj University. Madurai
M.E(MW &OPTICAL)	:A.C.College of Engg., & Technolgy, Karaikudi
PGDCA	: Madurai Kamaraj University, Madurai.

EXPERIENCE:

- 1991-1993 : Design Engineer, Priyadharshini Television(P) LTD, Karaikudi.
- 1993-1996 : Lecturer, Seethai Ammal Polytechnic College, Sivagangai.
- 1998-2003 : Lecturer, Arulmigu Kalasalingam College of Engineering.
- 2003-2007 : Research Scholar, Indian Institute of Technology, Chennai.
- 2007-2009 : Assistant Professor, RVSCollege of Engg., & Technology, Dindigul.
- 2009- Till Date : Kalasalingam University

DOMAIN KNOWLEDGE:

- Optical communication system,
- Microwave devices
- Linear integrated circuits

ACHIVEMENTS:

- Best trainer award for Radio & TV servicing under community polytechnic scheme.

Leading Birds

Name	Reg. No.	C.G.P.A
Final Year:		
P.Sriram	9908005327	9.54
J.PraiselineKarunya	9908005232	9.40
S.MohanaPriya	9908005179	9.00
Pre-final year:		
SajagSatayu Prasad Singh	9909005246	9.40
Chandankumar	9909005073	9.29
V.P.KeerthanaPriya	9909005137	9.12
Second year:		
PurasalaswarnaMadhuri	9910005073	9.50
MritunjayKumar	9910005116	9.00
Shilpee jaiswal	9910005089	8.89

GATE 2012 Scorers

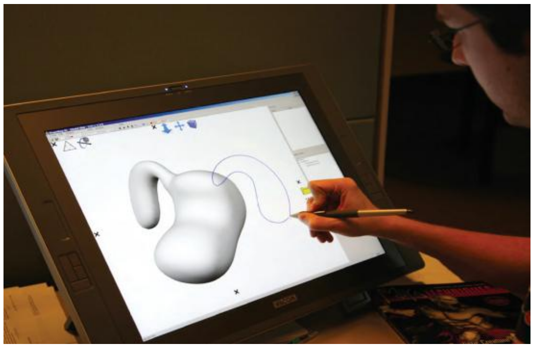
S.No.	Name	Percentile
1	NangineediManikanta	91.00
2	S. Deivanayagi	89.75
3	Indhumathy.V	86.00
4	Escaline Surya	84.83
5	ChavaliSaikartheek	84.00
6	Devarapalliprasath	84.00
7	M.Selvavasanth	83.00
8	Ravi Prakash Sati	81.19
9	S. Induja	80.15
10	Arjun R. Palaniappan	80.14
11	S. Nagarajan	80.00

12	Vijay Rajeshwarsharma.T	77.80
13	C. Arunkarthick	77.00
14	S. Sairamsubramanian	77.00
15	T.R.S.Rambalaji	75.39
16	E. Mahalakshmi	73.90
17	Siva shankar.V	71.08
18	J. Karthick	71.00
19	S. Abinesh	70.00
20	S. Ashok kumar	70.00
21	M. Arunkumar	70.00
22	S. Bhuvaneshwaran	70.00

Magazine Crew Members

TECH-TREATISE:

Pen Based Computing:



The heart of this vision was that the pen would remove the requirement for typing skills in order to operate a computer. Instead of typing, a user would simply write or draw, and the computer would recognize and act upon this input. The rationale was that by supporting this “natural” expression, computing would be accessible to everyone, usable in broad range of tasks from grandmothers entering recipes, to mathematicians solving problems with the aid of a computer. Like many visions of the future, this one was inspiring but not wholly accurate.

Certainly some of the key technologies to enable pen-based computing have come into fruition and have been adopted widely. However, the dream of ubiquitous handwriting and drawing recognition has not materialized. One can argue that this type of technology has yet to mature but will in the future.

What’s fascinating about pen-based computing is how it is being used in alternative ways from the original vision, which was only a slice of the rich variety of ways a pen can be used in human-computer interaction.

Pen, the Great Note-Taker:

Pen input research has also focused on the pen’s ability to be used to fluidly switch between text input, drawing and pointing.

Inspired by how people combine both drawing and handwriting in paper notebooks, these types of systems attempt to recreate and amplify this experience in the digital world. Ken Hinckley’s InkSeine application is a prime example. With InkSeine a user can quickly throw together notes where ink, clippings, links to documents and web pages, and queries persist in-place. Unlike a mouse and keyboard system, where a user must switch between keyboard and mouse for text entry and pointing, InkSeine supports all these operations in a free form way, reminiscent of paper notebooks.

Pen Input 3D

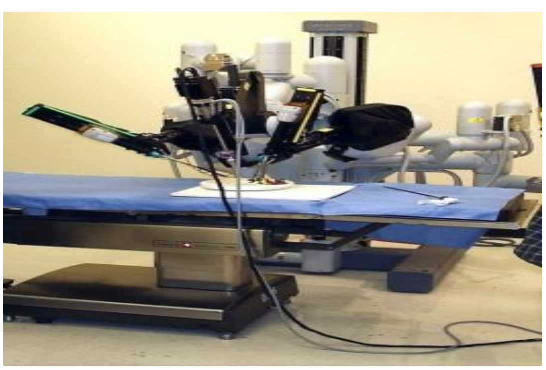
Pen input can also be used to create 3D graphics. A simple example is drawing a box in perspective and the computer automatically recognizes it as cube structure and creates the corresponding 3D geometry of a cube. The user can then rotate the object to see it from different viewpoints. In general, this is a computer vision problem—recognizing shapes and objects in the real world—and much progress on this general problem remains. However, researchers have made progress with systems where pen input is interpreted stroke by stroke into three-dimensional objects. Research systems such as I Love Sketch and Shape Shop allow drawing on planes in 3D space or directly on to 3D surfaces.

M.Bruntha
Pre final year

Satellite Surgery Robot Developed to Repair Satellites

Artificial satellites placed in orbits are used for a variety of purposes like communication, GPS, military applications, TV broadcasting, and so on. No matter how much time and money has been spent on building these hi-tech devices, satellites are basically just machines and constant use can damage any machine. Every year NASA is spending millions of dollars just for the maintenance of these satellites, but they have not found a feasible way to repair a damaged one.

Till present, there was only one option left for the researchers –completely blow–off the satellite!!They could not find any other feasible method to repair it. They tried using manpower by sending them directly to the repair site. This turned out to be a bad idea, as they could not adapt themselves to the outer space characteristics.



NASA had to face a lot of criticism for blowing up satellites in outer space. Since a single satellite contains millions of electronic equipment's, blowing them off would literally mean the disposal of e- waste all over the outer space. These toxic wastes could bring serious threats to humans by affecting the Earth's atmosphere in the coming future. A remotely operated robot could easily do the job but controlling of the robot will be a difficult task. Precise controlling of the robot is necessary, as things are different in outer space.

Engineers at John Hopkins University are known for designing robots for medical purposes. Currently they are working on a project in which a medical surgery robot called Da Vinci console is transformed into a robot that can be used for repairing satellites.

A brief description of this robotic surgery project has been presented by two graduate students Tian Xia and Jonathan Bohren from John Hopkins' Homewood campus. In this demonstration, researchers showed how the Da Vinci console, which was used for the treatment of cancer was modified as a satellite-repairing robot. It consists of a 3D eyepiece by which the operator can remotely control the robot. By using the 3D eye piece the user gets a better vision which helps him to repair with ease .It also includes a touch or haptic feedback to the operator.

This robot was successfully tested on 29th November 2011. The modified da Vinci console was used to control an Industrial robot at NASA's Goddard Space Flight Centre about 30 miles away. This test was successful and researchers concluded that they could use it in space mission too. The most challenging problem is that the satellites are travelling in an orbit and thus, prone to time delay for signal reception. Researchers may have used special algorithms to refine this problem. The robot can also be used for refueling satellites.

B.Prabhakar
Pre final year

The Brain Fingerprinting Technology

ABSTRACT:
Brain Fingerprinting is a new computer-based technology to identify the perpetrator of a crime accurately and scientifically by measuring brain-wave responses to crime-relevant words or pictures presented on a computer screen. Brain Fingerprinting has proven 100% accurate in over 120 tests, including tests on FBI agents, tests for a US intelligence agency and for the US Navy, and tests on real-life situations including felony crimes.

Why Brain Fingerprinting?

Brain Fingerprinting is based on the principle that the brain is central to all human acts. In a criminal act, there may or may not be many kinds of peripheral evidence, but the brain is always there, planning, executing, and recording the crime. The fundamental difference between a perpetrator and a falsely accused, innocent person is that the perpetrator, having committed the crime, has the details of the crime stored in his brain, and the innocent suspect does not. This is what Brain Fingerprinting detects scientifically.

Matching evidence at the crime scene with evidence in the brain

When a crime is committed, a record is stored in the brain of the perpetrator. Brain Fingerprinting provides a means to objectively and scientifically connect evidence from the crime scene with evidence stored in the brain. (This is similar to the process of connecting DNA samples from the perpetrator with biological evidence found at the scene of the crime; only the evidence evaluated by Brain Fingerprinting is evidence stored in the brain.) Brain Fingerprinting measures electrical brain activity in response to crime-relevant words or pictures presented on a computer screen, and reveals a brain MERMER (memory and encoding related multifaceted electroencephalographic response) when, and only when, the evidence stored in the brain matches the evidence from the crime scene. Thus, the guilty can be identified and the innocent can be cleared in an accurate, scientific, objective, non-invasive, non-stressful, and non-testimonial manner

Scientific Procedure

Brain Fingerprinting incorporates the following procedure. A sequence of words or pictures is presented on a video monitor under computer control. Each stimulus appears for a fraction of a second. Three types of stimuli are presented: "targets," "irrelevant," and "probes."

The targets are made relevant and noteworthy to all subjects: the subject is given a list of the target stimuli and instructed to press a particular button in response to targets, and to press another button in response to all other stimuli. Since the targets are noteworthy for the subject, they elicit a MERMER.

Some of the non-target stimuli are relevant to the crime or situation under investigation. These relevant stimuli are referred to as probes. For a subject who has committed the crime, the probes are noteworthy due to his knowledge of the details of the crime, and therefore probes elicit a brain MERMER. For an innocent subject lacking this detailed knowledge of the crime, the probes are indistinguishable from the irrelevant stimuli. For such a subject, the probes are not noteworthy, and thus probes do not elicit a MERMER.

Computer Controlled

The entire Brain Fingerprinting System is under computer control, including presentation of the stimuli and recording of electrical brain activity, as well as a mathematical data analysis algorithm that compares the responses to the three types of stimuli and produces a determination of "information present" ("guilty") or "information absent" ("innocent"), and a statistical confidence level for this determination. At no time during the testing and data analysis do any biases and interpretations of a system expert affect the stimulus presentation or brain responses.

The devices used in brain fingerprinting
Brain waves: Using brain waves to detect guilt.

How it works?A Suspect is tested by looking at three kinds of information represented by Different colored lines:
Red: information the suspect is expected to know
Green: information not known to suspect
Blue: information of the crime that only perpetrator would know

Case 1: Because the blue and green, Lines closely correlate, suspect does Not have critical knowledge of the crime.

Case 2: Because the blue and red, Lines closely correlate, and suspect has Critical knowledge of the crime.

From this methodology we can easily identify the culprits who are related to the crime. Now a day's Indian crime agencies also follow this methodology.

S.Sivasethuramalingam
V.G.Srinath
Pre final year

Space Shuttle program (1981-2011)

Discovery liftoff, 2008

The Space Shuttle became the major focus of NASA in the late 1970s and the 1980s. Planned as a frequently launch able and mostly reusable vehicle, four space shuttle orbiters were built by 1985. The first to launch, Columbia, did so on April 12, 1981 the 20th anniversary of the first space flight by Yuri Gagarin.
Its major components were a rocket plane orbiter with an external fuel tank and two solid fuel launch rockets at its side. The external tank, which was bigger than the spacecraft itself, was the only component that was not reused. The shuttle could orbit in altitudes of 185 – 643 km (115 – 400 miles) and carry a maximum payload (to low orbit) of 24,400 kg (54,000 lb.). Missions could last from 5 to 17 days and crews could be from 2 to 8 astronauts. On 20 missions 1983–1998 the Space Shuttle

carried Spacelab a space laboratory designed in cooperation with ESA. It was never deployed into orbit but was mounted to the Space Shuttle and went up and down with it on each mission. Another famous line of missions were the launch and later successful repair of the Hubble space telescope 1990 and 1993

In 1995 Russian-American interaction resumed with the Shuttle-Mir missions (1995–1998). Once more an American vehicle docked with a Russian craft, this time a full-fledged space station. This cooperation has continued to 2011, with Russia and the United States the two biggest partners in the largest space station ever built: the International Space Station (ISS). The strength of their cooperation on this project was even more evident when NASA began relying on Russian launch vehicles to service the ISS during the two-year grounding of the shuttle fleet following the 2003 Space Shuttle Columbia disaster.

The shuttle fleet lost two orbiters and 14 astronauts in two disasters: Challenger in 1986, and Columbia in 2003. While the 1986 loss was mitigated by building the Space Shuttle Endeavour from replacement parts, NASA did not build another orbiter to replace the second loss. NASA's Space Shuttle program had 135 missions when the program ended with the successful landing of the Space Shuttle Atlantis at the Kennedy Space Center on July 21, 2011.

The ISS combines the Japanese Kibō laboratory with three space station projects, the Soviet/Russian Mir-2, the American Freedom, and the European Columbus. Budget constraints led to the merger of these projects into a single multi-national programed in the 1990s. The station consists of pressurized modules, external trusses, solar arrays and other components, which have been launched by Russian Proton and Soyuz rockets together with the American Space Shuttles. It is currently being assembled in Low Earth Orbit. The construction began in 1998 and construction of USOS was completed in 2011; operations are expected to continue until at least 2020. The station can be seen from the Earth with the naked eye and, as of 2011, is the largest artificial satellite in Earth orbit with a mass and volume larger than that of any previous space station.

As mentioned it is a joint project between the five participating space agencies, NASA, the Russian RKA, the Japanese JAXA, the European ESA, and the Canadian CSA. The ownership and use of the space station is established in intergovernmental treaties and agreements which divide the station into two areas and allow Russia to retain full ownership of the Russian Segment, with the US Segment allocated between the other international partners. The station is serviced by Soyuz and Progress spacecraft's together with the Automated Transfer Vehicle and the H-II Transfer Vehicle and has been visited by astronauts and cosmonauts from 15 different nations. The Space Shuttle, before its retirement was also used for cargo and crew transfer.

In 2012, it is expected that the American Commercial Resupply Service vehicles, the SpaceXDragon and the Orbital

Sciences' Cygnus will begin to service the space station with cargo.

On February 1st, 2010, the American Commercial Crew Development program was initiated with the purpose of creating commercially operated spacecraft capable of delivering crew to the ISS. These spacecraft are expected to become operational in the mid-2010s.

Unmanned missions (1958)

Deep space mission deployed by shuttle, 1989

The first mission was Explorer 1, which started as an ABMA/JPL project during the early space race. It was launched in January 1958, two month after Sputnik. At the creation of NASA it was transferred to this agency and still continues to this day (2011). Its missions have been focusing on the Earth and the Sun measuring among others magnetic fields and solar wind

The closest planets Mars, Venus and Mercury have been the goal of at least 4 programs. The first was Mariner in the 1960s and '70s, which visited all three of them. Mariner was also the first to make a planetary flyby, to take the first pictures from another planet, the first planetary orbiter, and the first to make a gravity assist maneuver. This is a technique where the satellite takes advantage of the gravity and velocity of planets to reach its destination.

The first successful landing on Mars was made by Viking I in 1976. 20 years later a rover was landed on Mars by Mars Pathfinder.

Outside Mars, Jupiter was first visited by Pioneer 10 in 1973. More than 20 years later Galileo sent a probe into the planets' atmosphere. The first spacecraft to leave the solar system was Pioneer 10 in 1983. At a time it was the most distant spacecraft, but it is now passed by Voyager 2.

A problem with far space travel is communication. For instance, at present it takes about 3 hours for a radio signal to reach the New Horizon spacecraft at a point more than half way to Pluto.

On 26 November 2011, NASA's Mars Science Laboratory mission was successfully launched for Mars. The mission is scheduled to land the robotic "Curiosity" rover on the surface of Mars in August 2012, whereupon the rover will search for evidence of past or present life on Mars.

Recent and planned activities

NASA's ongoing investigations include in-depth surveys of Mars and Saturn and studies of the Earth and the Sun. Other active spacecraft missions are MESSENGER for Mercury, New Horizons (for Jupiter, Pluto, and beyond), and Dawn for the asteroid belt. NASA continued to support in situ exploration beyond the asteroid belt, including Pioneer and Voyager traverses into the unexplored trans-Pluto region, and Gas Giant orbiters Galileo (1989–2003), Cassini, and Juno (2011–).

The New Horizons mission to Pluto was launched in 2006 and aiming for Pluto flyby in 2015. The probe received a gravity assist from Jupiter in February 2007, examining some of Jupiter's inner moons and testing on-board instruments during the fly-by. On the horizon of NASA's plans is the MAVEN spacecraft as part of the Mars Scout Program to study the atmosphere of Mars.

On December 4, 2006, NASA announced it was planning a permanent moon base. The goal was to start building the moon base by 2020, and by 2024, has a fully functional base that would allow for crew rotations and in-situ resource utilization. However in 2009, the Augustine Committee found the program to be on an "unsustainable trajectory." In 2010, President Barack Obama halted existing plans, including the Moon base, and directed a generic focus on manned missions to asteroids and Mars, as well as extending support for the International Space Station.

On April 18, 2011, NASA awarded nearly $270 million to four companies to develop U.S. vehicles that could fly astronauts to the ISS as part of the CCDEV 2 program.

In September 2011, NASA announced the start of the Space Launch System program to develop a human-rated heavy lift vehicle. The Space Launch System is intended to launch the Orion Multi-Purpose Crew Vehicle and other elements towards the Moon, a near-Earth asteroids and one day Mars. The Orion MPCV is planned for an unmanned test launch on a Delta IV Heavy rocket around late 2013.

Vanishree. A
Pre final year

Difference between CCD and CMOS image sensors in a digital camera

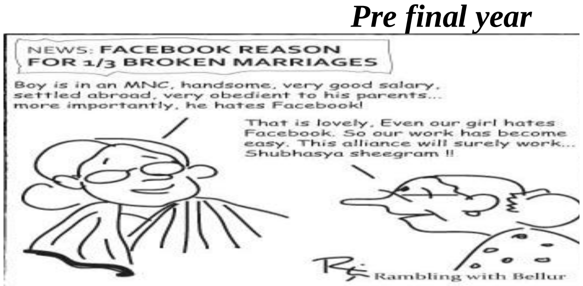
Digital camerashave become extremely common as the prices have come down. One of the drivers behind the falling prices has been the introduction of CMOS image sensors. CMOS sensors are much less expensive to manufacture than CCD sensors.

Both CCD (charge-coupled device) and CMOS (complementary metal-oxide semiconductor) image sensors start at the same point -- they have to convert light into electrons. If you have read the article How Solar Cells Work, you understand one technology that is used to perform the conversion. One simplified way to think about the sensor used in a digital camera (or camcorder) is to think of it as having a 2-D array of thousands or millions of tiny solar cells, each of which transforms the lightfrom one small portion of the image into electrons. Both CCD and CMOS devices perform this task using a variety of technologies.

The next step is to read the value (accumulated charge) of each cell in the image. In a CCD device, the charge is actually transported across the chip and read at one corner of the array. An analog-to-digital converter turns each pixel's value into a digital value. In most CMOS devices, there are several transistors at each pixel that amplify and move the charge using more traditional wires. The CMOS approach is more flexible because each pixel can be read individually. CCDs use a special manufacturing process to create the ability to transport charge across the chip without distortion. This process leads to very high-quality sensors in terms of fidelity and light sensitivity. CMOS chips, on the other hand, use traditional manufacturing processes to create the chip --the same processes used to make most microprocessors. Because of the manufacturing differences, there have been some noticeable differences between CCD and CMOS sensors.CCD sensors, as mentioned above, create high-quality, low-noise images. CMOS sensors, traditionally, are more susceptible to noise. Because each pixel on a CMOS sensor has several transistors located next to it, the light sensitivity of a CMOS chip tends to be lower. Many of the photons hitting the chip hit the transistors instead of the photodiode.CMOS traditionally consumes little power. Implementing a sensor in CMOS yields a low-power sensor.

CCD sensors have been mass produced for a longer period of time, so they are more mature. They tend to have higher quality and more pixels. Based on these differences, you can see that CCDs tend to be used in cameras that focus on high-quality images with lots of pixels and excellent light sensitivity. CMOS sensors traditionally have lower quality, lower resolution and lower sensitivity. CMOS sensors are just now improving to the point where they reach near parity with CCD devices in some applications. CMOS cameras are usually less expensive and have great battery life.

V.P.Kalyan Kumar
Pre final year



Robots- The future world

Robots play an ever increasing role in our world. A robot in fact an automatic machine, with electronic brains programmed to carry out specific tasks. Most robots are used in industry, for example nuclear scientists use robots to handle radioactive materials. In 1980s, also began to research the use of robots in routine medical surgery.

Isaac Asimov-Science Fiction Writer

ISAAC ASIMOV proposed three laws of robotics to allay the fears that one day robots could “take over”. These is as follows

- The robot must not harm people, or allow them to come to any harm
- Robots must obey their orders, unless this conflicts with the first law
- A robot must protect itself, unless this conflicts with other laws.

Science fiction robots

The first science fiction robot was in thePlay ROSSUM’S UNIVERSAL ROBOTS, by a Czech writer KAREL CAPEK in 1921.The theme of human like robots was developed in 1926 in film METROPOLIS

The comedy couple robots C3P0 and R2D2 from the star war films, where C3P0 is an android and R2D2 was to carry out repairs on spacecraft.

How a robot works?

A typical industrial robot is one armed machine with flexible joints equivalent to human shoulder, elbow and wrist. It has a gripping mechanism that works as a hand. The robotic arm swivels on its supporting base.May be moved electrically and pneumatically, by using compressed air. All the movements are controlled by the robot’s computer brain.

Uses of robots

Robots are often used in difficult or dangerous situations to carry out tasks that people wish to avoid.

- Many factories use robots in production lines. The car industry is the major user of robots for welding car components and spraying paints. The robot is programmed to carry out its tasks quickly and accurately.
- Robots are also used by security forces in bomb disposal operations. A mobile robot is used for bomb disposal. It is radio controlled. It has TV cameras and an adjustable arm with a grab attached for gripping.
- Space probes are the robots used to explore other planets. In 1976 American scientists sent two Viking probes to carry out a study of planet mars.
- Robots in home are used for performing simple, repetitive tasks that people would find it boring such as washing up cleaning. Research is being carried out to make more sophisticated robots that have independent movement and careful coordination of 'mind', 'eyes' and 'hands'.

The future of robotics

As the research into robotics continues more versatile and user friendly robots are being developed. Three dimensional vision and increased sensitivity enable industrial robots to carry out more routine jobs, while the advances in the artificial intelligence will give the robots more independence to solve the problems as they arise.

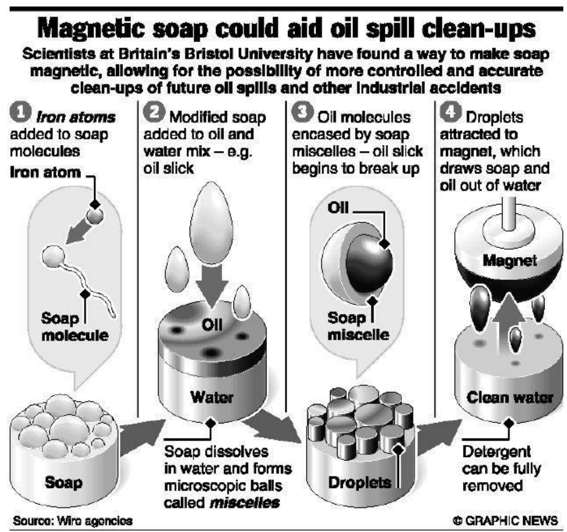
Artificial intelligence

The aim of artificial intelligence is to develop machines that can think and learn and interact with humans without having to be pre-programmed.

Let us all step into the world of robotics and come out with flying colors. "Winners don't do different things, they do things differently".

Sowmyapoorani
Pre final year

World's first magnetic soap produced



In a pioneering research, scientists claim to have produced the world's first magnetic soap that is composed of iron-rich salts dissolved in waterway team at Bristol University says that its soap, which responds to a magnetic field when placed in solution, would calm all concerns over the use of surfactants in oil-spill clean-ups and revolutionize industrial cleaning products. For long, researchers have been searching for a way to control soaps (or surfactants as they are known in industry) once they are in solution to increase the ability to dissolve oils in water and then remove them from a system. The Bristol University team produced the magnetic soap by dissolving iron in a range of inert surfactant materials composed of chloride and bromide ions, very similar to those found in everyday mouthwash or fabric conditioner.

The addition of the iron creates metallic centres within the soap particles, say the scientists led by Julian Eastoe. To test its properties, the team introduced a magnet to a test tube containing their new soap lying beneath a less dense organic solution, the *AngewandteChemie* journal reported. When the magnet was introduced the iron-rich soap overcame both gravity and surface tension between the water and oil, to levitate through the organic solvent and reach the source of magnetic energy, proving its magnetic properties. Once the surfactant was developed and shown to be magnetic, the scientists took it to Institute Laue-Langevin (ILL), the world's flagship centre for neutron science, to investigate the science behind its remarkable property.

When surfactants are added to water they are known to form tiny clumps (particles called micelles). At ILL, the scientists used a technique called “small angle neutron scattering (SANS)” to confirm that it was this clumping of the iron-rich surfactant that brought about its magnetic properties. From a commercial point of view, these exact liquids aren't yet ready to appear in any household product, by proving that magnetic soaps can be developed, future work can reproduce the same phenomenon in more commercially viable liquids for a range of applications from water treatment to industrial cleaning products.

Roshitachandel
Pre final year

Gesture recognition

A child being sensed by a simple gesture recognition algorithm detecting hand location and movement

Gesture recognition is a topic in computer science and language technology with the goal of interpreting human gestures via mathematical algorithms. Gestures can originate from any bodily motion or state but commonly originate from the face or hand. Current focuses in the field include emotion recognition from the face and hand gesture recognition. Many approaches have been made using cameras and computer vision algorithms to interpret sign language. However, the identification and recognition of posture, gait, proxemics, and human behaviors is also the subject of gesture recognition techniques.

Gesture recognition can be seen as a way for computers to begin to understand human body language, this building a richer bridge between machines and humans than primitive text user interfaces or even GUIs (graphical user interfaces), which still limit the majority of input to keyboard and mouse. Gesture recognition enables humans to interface with the machine (HMI) and interact naturally without any mechanical devices. Using the concept of gesture recognition, it is possible to point a finger at the computer screen so that the cursor will move accordingly. This could potentially make conventional input devices such as mouse, keyboards and even touch-screens redundant. Gesture recognition can be conducted with techniques from computer vision and processing. The literature includes ongoing work in the computer vision field on capturing gestures or more general human pose and movements by cameras connected to a computer.

Gesture recognition and pen computing:

In some literature], the term gesture recognition has been used to refer more narrowly to non-text-input handwriting symbols, such as inking on a graphics tablet, multi-touch gestures, and mouse gesture recognition. This is computer interaction through the drawing of symbols with a pointing device cursor.

Gesture types

In computer interfaces, two types of gestures are distinguished:

Offline gestures: Those gestures that are processed after the user interaction with the object. An example is the gesture to activate a menu.

Online gestures: Direct manipulation gestures. They are used to scale or rotate a tangible object.

Uses

Gesture recognition is useful for processing information from humans which is not conveyed through speech or type. As well, there are various types of gestures which can be identified by computers.

For socially assistive robotics. By using proper sensors (accelerometers and gyros) worn on the body of a patient and by reading the values from those sensors, robots can assist in patient rehabilitation. The best example can be stroke rehabilitation.

Directional indication through pointing. Pointing has a very specific purpose in our] society, to reference an object or location based on its position relative to ourselves. The use of gesture recognition to determine where a person is pointing is useful for identifying the context of statements or instructions. This application is of particular interest in the field of robotics.

Control through facial gestures. Controlling a computer through facial gestures is a useful application of gesture recognition for users who may not physically be able to use a mouse or keyboard. Eye tracking in particular may be of use for controlling cursor motion or focusing on elements of a display.

Alternative computer interfaces. Foregoing the traditional keyboard and mouse setup to interact with a computer, strong gesture recognition could allow users to accomplish frequent or common tasks using hand or face gestures to a camera.

Immersive game technology. Gestures can be used to control interactions within video games to try and make the game player's experience more interactive or immersive.

Virtual controllers. For systems where the act of finding or acquiring a physical controller could require too much time, gestures can be used as an alternative control mechanism. Controlling secondary devices in a car or controlling a television set are examples of such usage.

Remote control. Through the use of gesture recognition, "remote control with the wave of a hand" of various devices is possible. The signal must not only indicate the desired response, but also which device to be controlled.

Input devices

The ability to track a person's movements and determine what gestures they may be performing can be achieved through various tools. Although there is a large amount of research done in image/video based gesture recognition, there is some variation within the tools and environments used between implementations.

Wired gloves. These can provide input to the computer about the position and rotation of the hands using magnetic or inertial tracking devices. Furthermore, some gloves can detect finger bending with a high degree of accuracy (5-10 degrees), or even provide haptic feedback to the user, which is a simulation of the sense of touch. The first commercially available hand-tracking glove-type device was the Data Glove, a glove-type device which could detect hand position, movement and finger bending. This uses fiber optic cables running down the back of the hand. Light pulses are created and when the fingers are bent, light leaks through small cracks and the loss is registered, giving an approximation of the hand pose.

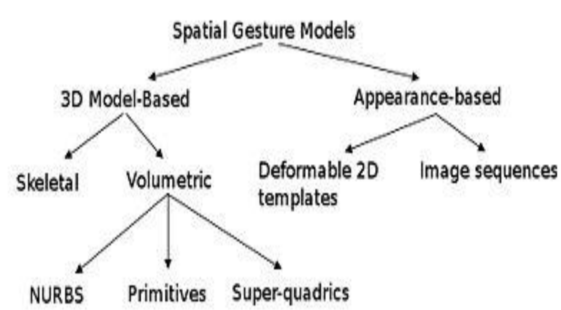
Depth-aware cameras. Using specialized cameras such as time-of-flight cameras, one can generate a depth map of what is being seen through the camera at a short range, and use this data to approximate a 3d representation of what is being seen. These can be effective for detection of hand gestures due to their short range capabilities.

Stereo cameras. Using two cameras whose relations to one another are known, a 3d representation can be approximated by the output of the cameras. To get the cameras' relations, one can use a positioning reference such as a lexian-stripe or infrared emitters. In combination with direct motion measurement (6D-Vision) gestures can directly be detected.

Controller-based gestures. These controllers act as an extension of the body so that when gestures are performed, some of their motion can be conveniently captured by software. Mouse gestures are one such example, where the motion of the mouse is correlated to a symbol being drawn by a person's hand, as is the Wii Remote, which can study changes in acceleration over time to represent gestures. Devices such as the LG Electronics Magic Wand, the Loop and the Scoop use Hillcrest Labs' Free space technology, which uses MEMS accelerometers, gyroscopes and other sensors to translate gestures into cursor movement. The software also compensates for human tremor and inadvertent movement.

Single camera. A normal camera can be used for gesture recognition where the resources/environment would not be convenient for other forms of image-based recognition. Although not necessarily as effective as stereo or depth aware cameras, using a single camera allows a greater possibility of accessibility to a wider audience.

ALGORITHMS

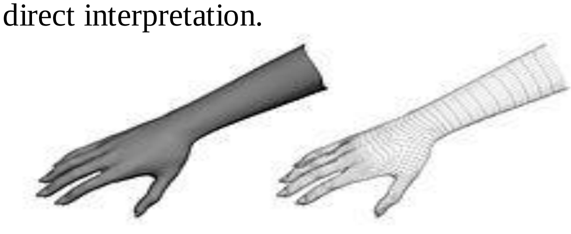


Different ways of tracking and analyzing gestures exist, and some basic layout is given is in the diagram above. For example, volumetric models convey the necessary information required for an elaborate analysis, however they prove to be very intensive in terms of computational power and require further technological developments in order to be implemented for real-time analysis. On the other hand, appearance-based models are easier to process but usually lack the generality required for Human-Computer Interaction.

Depending on the type of the input data, the approach for interpreting a gesture could be done in different ways. However, most of the techniques rely on key pointers represented in a 3D coordinate system. Based on the relative motion of these, the gesture can be detected with a high accuracy, depending of the quality of the input and the algorithm’s approach.

In order to interpret movements of the body, one has to classify them according to common properties and the message the movements may express. For example, in sign language each gesture represents a word or phrase. The taxonomy that seems very appropriate for Human-Computer Interaction has been proposed by Quek in “Toward a Vision-Based Hand Gesture Interface”. He presents several interactive gesture systems in order to capture the whole space of the gestures: 1. Manipulative; 2. Semaphoric; 3. Conversational.

Some literature differentiates 2 different approaches in gesture recognition: a 3D model based and an appearance-based. The foremost method makes use of 3D information of key elements of the body parts in order to obtain several important parameters, like palm position or joint angles. On the other hand, Appearance-based systems use images or videos for direct interpretation.

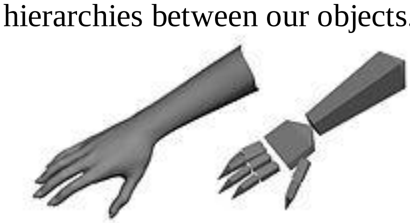


A read hand (left) is interpreted as a collection of vertices and lines in the 3D mesh version (right), and the software uses their relative position and interaction in order to infer the gesture.

3D model-based algorithms

The 3D model approach can use volumetric or skeletal models, or even a combination of the two. Volumetric approaches have been heavily used in computer animation industry and for computer vision purposes. The models are generally created of complicated 3D surfaces, like NURBS or polygon meshes.

The drawback of this method is that is very computational intensive and systems for live analysis is still to be developed. For the moment, a more interesting approach would be to map simple primitive objects to the person’s most important body parts (for example cylinders for the arms and neck, sphere for the head) and analyze the way these interact with each other. Furthermore, some abstract structures like super-quadrics and generalized cylinders may be even more suitable for approximating the body parts. Very exciting about this approach is that the parameters for these objects are quite simple. In order to better model the relation between these, we make use of constraints and hierarchies between our objects.



The skeletal version (right) is effectively modeling the hand (left). This has less parameter than the volumetric version and it's easier to compute, making it suitable for real-time gesture analysis systems.

Skeletal-based algorithms

Instead of using intensive processing of the 3D models and dealing with a lot of parameters, one can just use a simplified version of joint angle parameters along with segment lengths. This is known as a skeletal representation of the body, where a virtual skeleton of the person is computed and parts of the body are mapped to certain segments. The analysis here is done using the position and orientation of these segments and the relation between each one of them (for example the angle between the joints and the relative position or orientation)

Advantages of using skeletal models:

Algorithms are faster because only key parameters are analyzed.

Pattern matching against a template database is possible

Using key points allows the detection program to focus on the significant parts of the body

These binary silhouette (left) or contour (right) images represent typical input for appearance-based algorithms. They are compared with different hand templates and if they match, the correspondent gesture is inferred.

Appearance-based models

These models don't use a spatial representation of the body anymore, because they derive the parameters directly from the images or videos using a template database. Some are based on the deformable 2D templates of the human parts of the body, particularly hands. Deformable templates are sets of points on the outline of an object, used as interpolation nodes for the object's outline approximation. One of the simplest interpolation functions is linear, which performs an average shape from point sets, point variability parameters and external deformations. These template-based models are mostly used for hand-tracking, but could also be of use for simple gesture classification.

A second approach in gesture detecting using appearance-based models uses image sequences as gesture templates. Parameters for this method are either the images themselves, or certain features derived from these. Most of the time, only one (monoscopic) or two (stereoscopic) views are used.

Challenges

There are many challenges associated with the accuracy and usefulness of gesture recognition software. For image-based gesture recognition there are limitations on the equipment used and image noise. Images or video may not be under consistent lighting, or in the same location. Items in the background or distinct features of the users may make recognition more difficult.

The variety of implementations for image-based gesture recognition may also cause issue for viability of the technology to general usage. For example, an algorithm calibrated for one camera may not work for a different camera. The amount of background noise also causes tracking and recognition difficulties, especially when occlusions (partial and full) occur. Furthermore, the distance from the camera, and the camera's resolution and quality, also cause variations in recognition accuracy.

"Gorilla arm"

"Gorilla arm" was a side-effect of vertically-oriented touch-screen or light-pen use. In periods of prolonged use, users' arms began to feel fatigue and/or discomfort. This effect contributed to the decline of touch-screen input despite initial popularity in the 1980s.

Gorilla arm is not a problem for short-term use, since they only involve brief interactions which do not last long enough to cause gorilla arm.

S.Prasanna
Pre final year

Croma offers Intel Anti-Theft Service with Sandy Bridge-based laptops

Croma has announced it is now retailing Intel Anti-Theft Technology (Intel AT), as an optional upgrade for Sandy Bridge-based laptops at its stores. The initiative is said to be the first of it’s in kind in the country, and for now, is available exclusively at Croma.The Intel Anti-Theft Service is custom activated on laptops with second generation Intel Core processors, and allows customers to remotely lock down their missing, or stolen laptops. The technology, built-in, works even if the hard-drive is reimaged, a new hard-drive installed, or the boot order changed. The laptop will remain locked, and will not boot. The lock down is activated via an internet notification, or intelligent hardware timers.

Users will also be able to designate a space on the laptop’s hard drive as the secure data vault, where they can keep their confidential information.Users can also display a custom message on their locked laptops that might help facilitate a return by in case it is found when missing. Laptops can then be fully revived to normal functionality with the user’s personal password. Croma will be offering the theft protection service bundled in with select laptops as well as offering it as a paid option starting INR 199 for a period of 2 years from the day the user activates the Service.

V.G.Srinath
Pre final year

Photoshop

Introduction:

Adobe Photoshop is a graphics editing program developed and published by Adobe Systems Incorporated. Adobe’s 2003 "Creative Suite" rebranding led to Adobe Photoshop 8's renaming to Adobe Photoshop CS. Thus, Adobe Photoshop CS5 is the 12th major release of Adobe Photoshop. The CS rebranding also resulted in Adobe offering numerous software packages containing multiple Adobe programs for a reduced price. Adobe Photoshop is released in two editions: Adobe Photoshop, and Adobe Photoshop Extended, with the Extended having extra 3D image creation, motion graphics editing, and advanced image analysis features. Adobe Photoshop Extended is included in all of Adobe's Creative Suite offerings except Design Standard, which includes the Adobe Photoshop edition. Alongside Photoshop and Photoshop Extended, Adobe also publishes Photoshop.

Elements and Photoshop Light room, collectively called "The Adobe Photoshop Family". In 2008, Adobe released Adobe, a free web-based image editing tool to edit photos directly on blogs and social networking sites; in 2011 a version was released for the Android operating system and their OS operating system.

History:

In 1987, Thomas Knoll, a PhD student at the University of Michigan began writing a program on his Macintosh Plus to display greyscale images on a monochrome display. This program, called Display, caught the attention of his brother John Knoll, an Industrial Light & Magic employee, who recommended Thomas turn it into a fully-fledged image editing program. Thomas took a six month break from his studies in 1988 to collaborate with his brother on the program, which had been renamed ImagePro. Later that year, Thomas renamed his program Photoshop and worked out a short-term deal with scanner manufacturer Barneyscan to distribute copies of the program with a slide scanner; a "total of about 200 copies of Photoshop were shipped" this way During this time, John travelled to Silicon Valley and gave a demonstration of the program to engineers at Apple and Russell Brown, art director at Adobe. Both showings were successful, and Adobe decided to purchase the license to distribute in September 1988.

Features

Further information: Comparison of raster graphics editors**.** Photoshop uses color
models RGB, lab, CMYK, greyscale,
binary bitmap, and duotone. Photoshop has the ability to read and write raster and vector image formats such as .EPS, .PNG, .GIF, and.JPEG. Photoshop has ties with other Adobe software for media editing, animation, and authoring.

File Format:

The .PSD (Photoshop Document), Photoshop's native format, stores an image with support for most imaging options available in Photoshop. These include layers with masks, color spaces, ICC profiles, CMYK Mode (used for commercial printing), transparency, text, alpha channels and spot colors, clipping paths, and duotone settings. This is in contrast to many other file formats (e.g. .EPS or .GIF) that restrict content to provide streamlined, predictable functionality. PSD format is limited to a maximum height and width of 30,000 pixels. If it is then necessary to import the complex vector object into Photoshop, it is possible to import it into Photoshop as a Smart Object. When the Smart Object is double-clicked on Photoshop's layers palette, it is opened in its original software such as Adobe Illustrator where changes can be made, then after saving it, the Smart Object will be automatically updated in Photoshop.

P.Vengadesh
Pre final year

Cell that spoils shells: How to avoid mobile radiation

“From cell phones to computers, quality is improving and costs are shrinking as companies fight to offer the public the best product at the best price. But this philosophy is sadly missing from our health-care insurance system”. - John shadegg

In India, cell phone use has increased from 50 million to 800 million now and the number is increasing .The direct and indirect contributions of cell phones to the growth of the economy is unquestionable, but at what cost? This is a relevant question. The ill effects of radiation from Wi-Fi, laptops etc. are similar but mobile phones need to be looked at urgently because of increase in their usage. The radiation will not be from individual radios, television or defense transmitters, but practically every individual will act as a source of RF radiation. According to scientists researching the harmful effect of mobile radiation, there exists an electromagnetic wave pollution called as ‘electro smog’.

Evidence of emf's bio-effects:

Research an emf effects started at the time of world of war. It was noticed that birds fell from antenna towers on the warships. The birds not only died but were in a roasted, ready to eat condition. This stopped when radar power switched off. Also birds lost their direction whenever they reach antenna and fly n zigzag direction. The bird's ability to navigate with the help of emf and galactic field was known. It was identified that powerful emfs effect antenna radiated power confused the birds. Research thus started on harmful effects of powerful emf. It was found that the above certain level of EMF the cells in the body started split out.

The theory is that:

1. Continuous exposures to much lower level of RF radiation leads to serious health hazard.
2. Pulsed radiation s more powerful than continuous radiation.
3. Women are more susceptible to emf health hazard.

FCC regulation:

These limit the specific absorption rate to 1.6 watts per kilogram for part of the body (head exposure).0.08 watts per kg for whole body.4 watts per kg for hands, wrist. Based on the recommendation of IEEE 4 watt per kg is adopted for mobiles. Radio frequency has shown to induce heating of tissues leading to behavioral changes n mice, rats, and monkeys.

Recommendations to the government:

1. Immediately implement the 50 – meter safety zone around mobile towers.
2. Ensure that SAR rate is provided on front cover of cell phone.
3. Implement emf monitoring system near towers.
4. Advertise the harmful effects of mobiles to public

Caution for cell phone user:

1. Minimize the time you speak. Preferably use the mobile phone for short duration.
2. Avoid calling when the signal is weak, because the mobile then radiates maximally.
3. Avoid putting the cell phone in your left packet because your heart is closer.
4. Buy cell phones with low SAR rating.
5. Use a rear shield for the cell phone to prevent radiations towards your body.
6. Keep the cell phone away while you sleep.

R.Gowsaliya
Pre final year

Technical Information

Electronic Pills - Collecting Data inside the Body

After years of investment and development, wireless devices contained in swallow able capsules are now reaching the market. Companies such as Smart Pill based in Buffalo, New York and Israel-based Given Imaging (PillCam) market capsules the size of vitamin tablets that contain sensors or tiny cameras that collect information as they travel through the gastrointestinal tract before being excreted from the body a day or two. These new electronic inventions transmit information such as acidity, pressure and temperature levels or images of the esophagus and intestine to your doctor's computer for analysis. Doctors often use invasive methods such as catheters, endoscopic instruments or radioisotopes for collecting information about the digestive tract. So device companies have been developing easier, less intrusive ways, to gather information. Digestive diseases and disorders can include symptoms such as acid reflux, bloating, heartburn, abdominal pain, constipation, difficulty swallowing or loss of appetite. new electronic inventions "One of the main challenges is determining just what is happening in the stomach and intestines." says Dr. Anish A. Sheth, Director of the Gastrointestinal Motility Program at Yale-New Haven Hospital. Doctors can inspect the colon and peer into the stomach using endoscopic instruments. But some areas cannot be easily viewed, and finding out how muscles are working can be difficult. Electronic pills are being used to measure muscle contraction, ease of passage and other factors to reveal information unavailable in the past.

Transparent Electronics

Inventors, Jung Won Seo, Jae-Woo Park, Keong Su Lim, Ji-Hwan Yang and Sang Jung Kang, who are scientists at the Korean Advanced Institute of Science and Technology, have created the world's first transparent computer chip. new electronic inventions The chip, known as (TRRAM) or transparent resistive random access memory, is similar to existing chips known as (CMOS) or metal-oxide semiconductor memory, which we use in new electronic inventions. The difference is that TRRAM is completely clear and transparent. What is the benefit of having transparency? "It is a new milestone of transparent electronic systems," says Jung Won Seo. "By integrating TRRAM with other transparent electronic components, we can create total see-through embedded electronic systems." The technology could enable the windows or mirrors in your home to be used as computer monitors and television screens. This technology is expected to be available within 3 to 4 years.

Apple I Transparent Mobile Phone

Apple Phone - concept phone on one hand, clear conceptual phones already, so this is not just the first, but on the other, the so-called Window Phone has one impressive feature - its transparent housing varies depending on the weather! Thus, in the sunny days, the screen will be completely transparent, on a rainy day it will show virtual drops on the screen, and on a snowy day it is totally covered with frost, i.e. the translucent screen will look like as well as present a window into a variety of weather.

K.Abirami
Pre final year

Embed success Through Embedded - "The Golden Bird"

Embedded systems are everywhere today. These tiny electronics are present in anything electronics that seems intelligent like ATMs, MP3 players, TVs, Automobile's etc. They sit in your device to add an element of smartness in them.

What Is It?It is a combination of computer hardware and software. These system are designed to perform specific function in a given time frame-for ex:"to help you choose how long your washing machine runs, confer thinking power to microwave ovens, etc..

Market's Demand:-The India's semiconductor design industries employed a workforce of 160,000 in 2010, of which embedded accounts for as much as 82 per cent employment, thus highlighting the fact that embedded system open up a plethora of opportunities for their practitioners.

India's semiconductor consumption is to grow the fastest across the globe through 2015 at a compound annual growth rate of 15.9 per cent to nearly $15 billion. So with India becoming the hub for semiconductor design, the embedded industry offer tremendous job opportunities countrywide. We need tons of embedded professional to create all that is possible-be in telecom, networking, automotive, medical and so on.

Roles to Choose:-An embedded system is a team work of three different sets of people including hardware, software and application domain experts. You can look job under any of these categories. Hardware platform offer prototyping, debugging and testing. Software platform offers programmer with the responsibility of task like analysis and optimization of embedded software for the targeted real time operating system. Being in wireless engineer protocol domain you will be responsible for the development, integration and testing of various protocols within an embedded firmware stack used in mobile

Who is hiring:-Some of the major recruiters in this sector are Texas instruments, STMicroelectronics, Intel, Honeywell, Delphi Huawei, Mistral, LG electronics, Samsung, ATS, Sankhya technologies etc. So engineering students with their knowledge related to embedded system that their knowledge will become increasingly valuable to a number of different sectors including automobile, medical, consumer electronics, aerospace, military and much more.

Healthy Pocket:- The salaries are par with IT & ITs and in most cases slightly better. It is a highly paying field offering salaries roughly twice that of engineer in an enterprises field. Private company is always ready to offer loftier figure for candidates with suitable exposure.

AishwaryaGautham
Pre final year

Key loggers

Key loggers are one of the most well-known and feared security threats on computers today. They're feared because they are generally hard to detect, and because the damage they do is often meant to extend beyond the infected computer. A virus may seek to crash a computer, ruin its hard drive, or take some files, but a keylogger is usually employed to take personal information, be it a password or credit card number. There are many ways to protect against keyloggers' firewall is a great defense against key loggers because it will monitor your computer's activity more closely than you ever could. Upon detecting that a program is attempting to send data out, the firewall will ask for permission or display a warning.

S.Srikamatchi
Pre final year

Motion detectors

The term '**Motion detectors**' can be used to refer to any kind of sensing system which is used to detect motions; motion of any object or motion of human beings. However, it is primarily used to detect motion of human beings or in other words, presence of a body in a certain area.

A motion detector is an electronic device that detects the physical movement in a given area/ designated locations and it transforms motion into an electric signal. Applications to detect motion do so by either:

• Generating some stimulus and sensing its reflection.

• Sensing some signals generated by an object.

All **motion sensors** indicate the same thing; that some condition has changed. All sensors have some 'normal' state. Some sensors only report when the 'normal' status is disturbed, others also report when the condition reverts to 'normal'.

Motion sensors are commonly used in security systems as triggers for automatic lights or trips for remote alarms and similar applications. Motion sensors work based on a wide variety of principles and is used in a wide variety of applications. Typical usage could be in the exterior doorways or windows of a building for monitoring the area around the building. Upon detecting motion, they generate an electrical signal based on which some actions are taken. Some operate in much the same way as a military radar scanner, while others work based upon vibration, infrared radiation and even sound. All of these different types of motion sensors have different strengths and weaknesses, which are important to take into account when making a decision to choose a particle motion detection sensor.

TYPES OF MOTION SENSING

Motion sensors are employed to detect different types of human movement. Some are intended for local event sensing, some for area sensing

• Local Sensing

Local Sensing implies sensing of a motion at designated locations. Some of the motion sensors commonly used for this purpose are

- Visible/ Infrared light (LED/Laser) beam
- Contact Switch
- Piezoelectric Sensors
- Piezoresistive Sensors, etc.

• Area Sensing

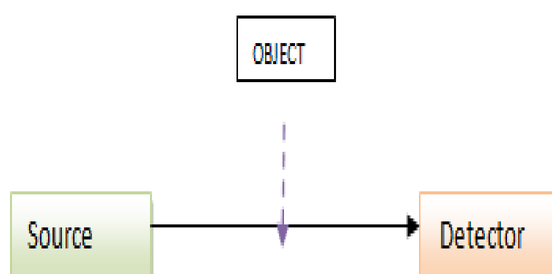
Area Sensing implies sensing of a motion in a specific 'Field of View (FOV)'. Motion sensors commonly employed for such application are

- Active/ Passive Infrared Motion Detector
- Ultrasound motion detection sensor
- Footstep Detection Sensing
- Microwave Doppler sensor

LOCAL MOTION SENSING

• Visible/ Infrared light (LED/Laser) beam based motion detectors

This kind of motion sensors employs break beam principle; in a light source transmits a beam of light towards a distant receiver creating an "electronic fence". Once a beam is broken/ interrupted due to some opaque object, output of detector changes and associated electronic circuitry takes actions.



• Contact Switch based motion detectors

Such switches can be placed under the mat and can be simplest of make-break types. In normal state, the switch is open, but when a person walks over the mat, the switches get closed. The status of the switch helps in detection of the movement.

• **Piezoelectric/ piezoresistive Sensors based motion detectors**

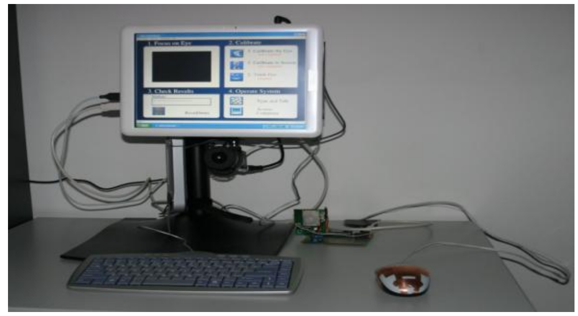
In place of contact switches, piezoelectric/piezoresistive sensors can be used. Piezoelectric sensors use piezoelectric effect. These generate electrical output on application of pressure. Piezoresistive sensors change their resistance on application of pressure. The change in resistance is measured to calculate applied pressure, i.e., movement over the sensor.

V.Vedhanarayanan
Prefinal Year

The ERICA Eye Tracker

The ERICA eye tracker is a low-budget commercial system of Eye Response Technologies and was used for all presented studies. The ERICA system shall serve as an example to present an eye tracker in detail.

The ERICA eye tracker is a system without head tracking and requires the fixation of the user's head with a chin rest. Although it provides some software for eye-gaze data analysis, it is a typical system for accessibility. The need of a chin rest does not matter for a disabled person who cannot move but changes the experience for a per-son without disabilities. It comes as a tablet PC with an external infrared camera, which both could be mounted at a wheel chair. For all studies the eye tracker was fixed.



The tablet PC has a standard PC architecture with a 1.4 GHz processor and 512 Megabyte RAM. The operating system installed is Windows XP SP2. As it uses a time slice of 10 milliseconds, 10 milliseconds are the best accuracy that can be achieved for any system event e.g. mouse moves. The TFT display can switch pixels with about 20 ms.

The ERICA system uses the white pupil method and a corneal reflection. The camera delivers the image of one eye with a resolution of 640 x 480 pixels at a frame rate of 60 Hz or about one frame every 17 milliseconds. The ERICA system reports the gaze position with the same rate in the case the eye does not move (during fixations). During eye movements the eye tracker does not deliver any data. This means there is a time gap in the data during a saccade.

Geometry of the Experimental Setup

The tablet PC's display has a size of 246 mm x 185 mm (12') and a resolution of 1024 x 768 pixels. The distance d of the eyes from the screen, given by the position of the chin rest, is 48 cm ± 2 cm. From $\tan(\alpha/2) = w/(2*d)$ the angle α calculates to $\alpha = 28.7°$. This means in average 0.028° per pixel or around 36 pixels for 1°. In the center of the screen, which is flat and not sphere shaped, the pixels per degree are about 35 pixels. As the dependency of the pixels per degree from the viewing angle is small, all calculations for the evaluation of the user studies use the average value of 36 pixels for 1°.

The ERICA-API uses the Windows message concept to deliver the data from the eye tracker to a window which has to process the message. The use of the API within Windows programming projects is easy and straightforward. After requesting a message name (ericaGetMessageName), registering the message at the Windows operating system (RegisterWindowMessage) and setting the receiver window (ericaSetReceiver), the ERICA system sends messages with the x-y position of the eye gaze and the x-y extend of the pupil in the message parameters (WPARAM, LPARAM).

The simplicity of the API is sufficient for many possible applications and makes it easy to start programming. There are some problems and bugs in the ERICA-API. The message stamp delivered with the ERICA messages is corrupt and this makes it necessary to use the system time (GetTickCount) at the time of message receipt instead of the time of message generation. This can influence the accuracy of timings but only if the system is very busy. Another problem is the delivery of coordinates relative to the client area of the window. It is an easy task to transform the coordinates into screen coordinates with a Windows API function (ClientToScreen). Moving the receiving window, however, causes a new gaze position relatively to the window and produces an ERICA message which appears as an eye movement.

If the receiving window is minimized, wrong coordinates can be reported. It also seems that occasional signed-unsigned mismatch errors occur when delivering coordinate values. Once aware of the problem one line of code can fix this.

From the view of a programmer, it is a little bit cumbersome that only one application at a time gets the ERICA messages. This makes it impossible to run two applications which use the ERICA-API in parallel. A lost-signal-notification in the case the image processing cannot recognize the eye, for example because it moved out of the focus of the camera, is also desirable. From the view of a scientist, it is a pity that there is no access to the calibration data and only one Boolean parameter (bWantSmoothedData) for controlling the filter. Without access to the calibration data it is not possible to program alternative calibration procedures or to adjust the calibration while using the eye tracker.

Without control of the filtering mechanism the existence of a minimum saccade length and strange effects as the gap in the data for saccade lengths have to be accepted. The fact that the ERICA system does not report data during a saccade, which is a property of the filter mechanism, inhibits the prediction of the saccade end point and time when the saccade movement is on half of the way.

The blinking detection (ericaDidBlinkOccur) worked fine, but requires a longer closing of the eye than for the natural blink. Therefore, the use of blinking does not bring a speed benefit compared to the dwell method. According to informal talks with test users blinking to trigger commands does not feel comfortable. Especially repetitive blinking over a longer time, as it is typical for computer interaction, seems to create a feeling of nervous eye muscles. Therefore, the conducted user studies did not use triggering of actions by blinking. The following relations apply to all distances given in pixels within this thesis. 1° = 36 pixels 1 pixel = 0.028°

The ERICA-API

A simple SDK (software development kit) to access the ERICA system on Windows platforms by programs is available. It provides four functions defined in the header file:

```
// used to tell ERICA where to send its messages
Void PASCAL ericaSetReceiver (HWND hwnd, BOOL bWantSmoothedData);
// used to get the name of the ERICA message
Void PASCAL ericaGetMessageName (char *pchName );
// Used to get the low and high words of a lVal
Void PASCAL GetSignedLowHigh (long lVal, int *pwLow, int *pwHigh );
// When processing an ERICA message, this is used to determine
// if a blink occurred just prior to the received data
BOOL PASCAL ericaDidBlinkOccur ();
```

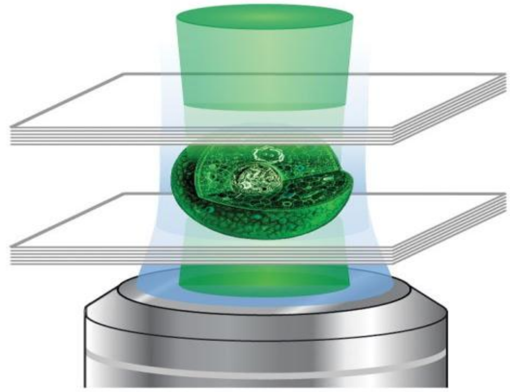
28 2 Overview and Related Work

B.Prabhakar,
Pre final year

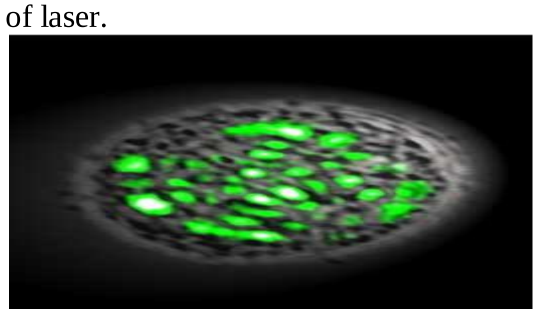
Fire Laser from Your Body

Cyclops, a superhero from the famous comics "X men" has been the favourite of many people across the world. In the story, the character produces powerful laser beams by optical blast from his eyes. This fiction has become a reality by the joint efforts of Seok-Hyun Yun and Malte Gather, two physicists at Massachusetts General Hospital.

They were able to develop a technique by which laser can be produced from human cells. The device used to produce this laser consists of the same components that are used to produce a real time laser. The pump source is used to produce the initial energy, the optical cavity to concentrate the energy, and the gain medium is a substance in which the electrons are exited until they go from a lower energy level to a higher energy level.



The researchers began experimenting on human kidney cells to produce green florescent proteins (GFP). This substance is known to produce luminance in jellyfish. They began by placing the modified cell between two mirrors there by creating an optical cavity (a cell sandwich as called by researchers). Then a blue light pulse was passed through the cell and the light bounced between the mirrors, causing the cell to glow. When the cell was observed through a microscope, they saw that the cell was glowing with a spot of laser.



Researchers suggest that this laser is suitable for many medical applications. For the diagnosis of diseases, nowadays a laser beam is passed into the body to get images or to attack the disease causing cells. There might also be a time when the human cells can be illuminated as laser. The illuminated cells can gather more information about the interior of the human body by penetrating the damaged tissues more deeply. More research has to be done to realize this as laser needs an external light source to illuminate. Some say that this technology will be more useful in gaining information about individual cell characteristics than medical applications as the laser needs an external light source and it is difficult to produce it inside human body. Yen suggests that by integrating a Nano scale cavity to laser cell we can produce a cell that will illuminate itself without any external source. Let's hope that this technology will come into reality in the near future itself – who doesn't want to shoot a laser from our own body!

B.Prabhakar
Pre-Final year

Robotics: a friend for assisting handicapped people

INTRODUCTION:

People with upper-limb impairments, including people with multiple sclerosis, spasticity, cerebral palsy, paraplegia, or stroke, depend on a personal assistantfor daily life situations and in the working environment. The robot arm MANUS was designed for such people, but they frequently cannot use it because of their disability. The control elements require flexibility and coordination in the hand that they do not have. To facilitate access to technological aids, control functionality of the aids has to be improved. Additionally, it should be possible to enter more abstract commands to simplify the use of the system. This has been partly realized in the commercially available HANDY system. The HANDY user can choose between five fixed actions, such as eating different kinds of food from a tray or being served a drink. However, the restricted flexibility of HANDY is a big disadvantage, as a fixed environment is a prerequisite.

To place the full capacity of technical systems like the robot arm MANUS with 6 degrees actions. Of freedom (DOF) at the user's disposal, a shared control structure is necessary. Low-level commands and taking advantage of all cognitive abilities of the user lead to full control flexibility. To relieve the user, semiautonomous sensor-based actions are also included in the control structure. Actions, such as gripping an object in an unstructured environment or pouring a cup with a drink and serving it to the disabled user, may be started by simple commands. The user is still responsible for decisions in situations where errors are possible, such as locating objects or planning a sequence of preprogrammed actions.

Hardware and software:

The Robotic system consists of an electric wheelchair (Merya, Germany), a MANUS robot arm (Exact Dynamics, Holland), and a dual Pentium PC, which is mounted in a special rigid box behind the wheelchair as shown.

The robot arm is linked to the PC via a CAN-bus to exchange commands and measurement data. The wheel-chair receives commands via RS-232 interface. A tray mounted on the left-side of the wheelchair and a flat LCD display as part of the MIMI completes the design.

To integrate the semiautonomous actions, the system includes cameras. Currently, a mini camera is mounted on the top of the gripper. For future applications, a stereo-system will be placed behind the user's head.

Architecture of the system FRIEND.

Preprogrammed actions, like pouring, may be generated off-line, using either a classical teach-in procedure or by robot programming by demonstration (RPD). RPD combines programmed movements with scripts that can be parameterized. A disadvantage of preprogrammed movements is that no variations are admissible. The item (the cup or bottle), its position, and initial conditions (the fluid level) must not be different compared to the programming situation. RPD may partially eliminate this disadvantage.

In an unstructured environment the use of preprogrammed actions is unsuitable. Therefore, FRIEND uses a combination of control by the user and autonomous control by the system. If, for instance, the user wants something to drink, he first will have to move the robot arm close to a bottle using speech commands like "Arm left" or "Arm up." As soon as the hand-mounted camera recognizes the object, the user initiates a pick action with the command "Pick!" In the pick action, the gripper is moved into an adequate position relative to the bottle using visual serving, and the bottle is gripped automatically.

An additional complication occurs if a desired object isn't located in the robot's workspace. A routine work problem is to remove a folder from a shelf. In this case, the wheelchair first has to move close to the shelf. When the wheelchair is in a suitable position, close enough to the shelf to reach the folder, the camera system has to recognize the folder. During this recognition process, the user supports the system with verbal commands to move the cameras close to the folder. When the folder is detected successfully, the command "Dock!" activates the docking action. A suitable wheelchair position is determined with the help of the imaginary links method to solve the inverse kinematic problem. In this case the wheelchair and the robot arm form a redundant system with 9 DOF. During every robot arm and wheelchair action, an intelligent part of the arm controller, the so-called Kinematics Configuration Controller (KCC), computes collision-free trajectories with additional sensor information.

Programming complex motions by demonstration:

It demonstrates the task to be executed with his own hand. The motions are measured, recorded, and process .To program new motions, the robotic system is equipped with two programming environments. In keyboard mode, the robot is programmed in the traditional way, available in almost all robots. In this mode, with the keyboard point-to-point positions on a trajectory are generated that are traced and stored in a database. In programming by demonstration mode (RPD), the progressed so that the robot can reproduce them. Many approaches described in the literature share a common feature they are designed mainly for simple pick-and-place applications like those found in industry, such as loading palettes and sorting and feeding parts. Neither the demonstrated motion trajectory nor the dynamics of the motion, such as the speed or general time response, are considered. But in the field of rehabilitation robots like FRIEND, where the tasks are much more complicated, this information is of great importance to enable the robot to execute these tasks correctly. Moreover, the methods mentioned don't consider that the human operator acts as a part of a closed loop shown below, acting as a sensor, control algorithm, and actuator. The loop is closed across the environment.

This closed loop enables a human being to execute even very complex trajectories without difficulty and in spite of variable initial conditions. For example, in the described scenario "pouring a glass from a bottle," the human continually observes the level of the glass and the flow from the bottle visually. With this information, he controls the motion of the bottle and fills the glass to a constant level with different shaped bottles and glasses, independent of the initial levels. To achieve similar robustness with a robot, a feedback loop must be installed.

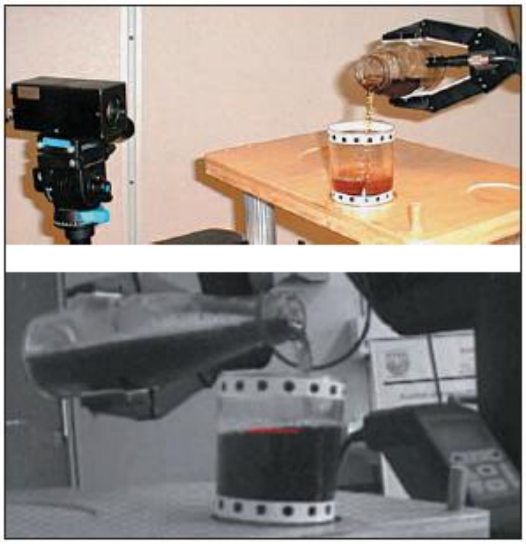
The demonstrated motions are divided into subsequences. The robot may repeat some as demonstrated, whereas others have to be part of the control loop. Additional sensor information is included and the controller will modify the recorded trajectory. In the next section we explain this method in detail for the "pouring a glass" scenario.

Trajectory generation:

Demonstration:

First, the trajectory for the pouring process has to be generated. To detect the motion of the bottle, a 6 DOF position sensor attached to the bottle and a transmitter, which represents the reference system for the motion detection, is used.

The programmer grasps an open bottle, places it in a relative pose to a glass, starts recording, and demonstrates the characteristic motion of this action. During the demonstration, the current pose of the bottle relative to the transmitter is sampled at regular time intervals. These pose data and the corresponding time represent the motion trajectory and are stored in an ordered list. With this detection method it is possible to precisely copy the demonstrated motion and its dynamics. The investigation of the pouring process yielded that it is sufficient to demonstrate one process, in which a full bottle is emptied completely. This overall trajectory contains all the movements for all initial levels.



The actual level of the fluid in the glass is determined by the camera and the flow from the bottle and the volume in the glass are calculated. The controller translates the difference between the desired and the computed flow to new offsets in the trajectory list and controls the flow in the way. Figure shows the volume in the glass, the computed flow from the bottle and control output.

Experimental Results

Robust positioning of the gripper:

Grasping an object with a support system like FRIEND in direct control mode is a time-consuming task for the user, even with the speech input device. On the other hand, the implementation of a general grasp utility leads to very complex algorithms. Since the number of frequently handled objects in our scenario is limited, a supporting utility that handles most of the necessary objects is already very helpful.

FRIEND uses a grasp utility for known labeled objects. Visual serving with a gripper-mounted camera is used in conjunction with teaching by showing (TbS). With TbS the gripper and the labeled object are placed in the desired relative position, the corresponding image is taken, and the desired features (as image features we use the centers of gravity of the four marker points) are extracted and stored.

Start and end of pick action.

If the same object has to be grasped again, it is necessary to return the gripper to the taught grasp position. This can be done either manually using basic spoken commands or automatically using visual serving with the gripper-mounted camera. In the latter case, the user has only to move the gripper into a position where the camera can detect the label on the object and to initiate the pick action. To support the user, a live camera image is represented on the flat screen. With visual serving, the gripper is moved to the position where the actual image features correspond to the taught

Features.

The visual serving controller computes the change in the position of the robot by comparing the measured and the desired image features and moves the robot to the calculated position. In the new position, another image is taken and the control algorithm is repeated until the taught relative position is reached. The robot is driven in Cartesian space with a look-and-move strategy. A controller suitable in a rehabilitation robotic system must:

Drive the gripper from any reasonable starting position to the gripping position

Must ensure that the object marker remains in the image during the motion.

Semi-automatic docking-action:

Positioning the wheelchair too close to the object causes collision with the shelf. Grasping without collision from a surface (optimal) wheelchair location. The obvious strategy of “the closer the better” is not always correct for a room which obstacles, as illustrated above. When the wheelchair is too close to the object the collision with the shelves occurs. Choosing a suitable location in a real environment manually is very arduous, since this location can only be determined during the grasping process. For users with limited head mobility, it is almost impossible to estimate a clustered spatial situation correctly.

A semi-autonomous Docking System assists the user. A suitable wheelchair location is calculated from spatial information about the objects and obstacles. To recognize the object and reconstruct the scene, stereo cameras are needed.

Experiences and future development:

After a short time of training the user is able to grip different kinds of objects and move them close to him. But some parts of these manipulation actions are very laborious; especially exact positioning, as they are needed to grasp an object, and the execution of a complex task, like pouring water from a bottle into a glass, are hard to handle.

The reasons are time delays in the speech recognition system and the tremendous amount of elementary commands that are needed to realize a complex task. Particularly, the last point isn't justifiable because of the long command sentences that had been introduced.

For security reasons, it requires the whole concentration of the user and is very tiresome. It is possible to shorten the command sentences, but in practice in a noisy environment, experience showed that these long sentences were necessary. As discussed, a marked improvement arises with the introduction of semiautonomous actions. In combination with the speech input system, this provides the user with a set of robust actions, which can be combined to accomplish complex tasks.

In spite of the theoretical and practical results presented above, a huge amount of research and development is still required. Therefore, future work will focus on the analysis and implementation of semiautonomous actions, the simplification of the MMI, and improvement of the safety functions.

SriVidhya

Pre final year

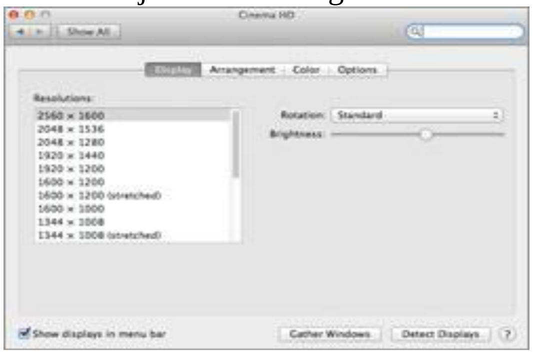
Tips To enhance your laptop Battery life

Here are top tips to enhance your battery throughput and also increase its overall lifespan:

- Charge your newly bought laptop fully and then discharge it up to 7% 2-3 times and always use your laptop like a mobile never use it by connecting it directly to an AC power outlet then your laptop will work fine even after years of use.
- Power management features: Our body feels energetic after a good sleep. The same goes for your laptop also. You can keep your laptop in "sleep" mode whenever you are not working. Use the 'power management' feature of 'windows' from the 'control panel' to utilize this feature.
- Screen brightness adjustment: The brightness level of the screen also decides the life of the battery. It is very common that we forget to adjust the brightness while working which consumes more energy. Set your brightness to a lower level when working on

battery backup to enhance the backup time & performance.

Adjust Screen Brightness.



At full brightness, the 13-inch Mac Book Air's screen measures an average of 285 lux. At 40-percent brightness (6 bars), its just 29 lux, but still very viewable. And brightness eats power.

To lower screen brightness:

Tap F1 repeatedly until the brightness is at a level just tolerable enough to view comfortably.

Alternatively:

Open System Preferences > Displays.

Adjust the Brightness slider to the level you're most comfortable with.

- Optimize Graphics: Do we actually require fancy graphic drivers all the time? The power utilized by graphics cards is almost equal to hard drives. Therefore you can understand that how much power they use.
- External Peripherals: Disconnect the peripherals like printers, scanners. USB drives etc. when not in use to save power.
- Check adapter voltage: Always check the voltage of your adaptor and make sure that you are not using AC adaptor with an incorrect voltage.
- Less utilization of CD and DVD drives: Do not keep a disk in the drive all the time. They utilize power which can decrease the life of battery.
- Remove battery: If you are not going to use your laptop for some days then it is always better to remove the battery from the laptop and store it separately.
- Disable Wireless card when not in use: Always disable your wireless card when it is not in use to avoid battery from draining out unnecessarily.
- Avoid undue heat: Do not keep your laptop in a closed car or under direct sunlight. It kills the life of the battery.
- **Turn off the auto save function**. MS-Word's and Excel's auto save functions are great but because they keep saving regular intervals, they work on your hard drive harder than it may have to. If you plan to do this, you may want to turn it back on as the battery runs low..
- Lower the graphics use. You can do this by changing the screen resolution and shutting off fancy graphic drivers. Graphics cards (video cards) use as much or more power today as hard disks

Practice the above tips in your day to day life to see your laptop battery serving you for at least 4-5 years...

V.Sidheswari
Pre final year

Samsung bada

Bada is a Linux based OS developed by Samsung for as they put it in Smartphone. Samsung announced it in the year 2010 BADA 1.0 when the OS was first launched it was a very basic OS with just the bare essential features to classify it as a Smartphone. With the upgrade BADA 2.0 is updated and it bring with speech recognition, multitasking, full-fledged web browser with flash support. It contains more than 7000 apps.samsung has announced plans to make this OS open source and to also integrate it into their upcoming smart TV as well.

S.Srikamatchi
Pre final year

Introduction to Computer Network Protocols

The word protocol is derived from the Greek word "protocol on" which means a leaf of paper glued to manuscript volume. In computer protocols means a set of rules, a communication language or set of standards between two or more computing devices. Protocols exist at the several levels of the OSI (open system interconnectivity) layers model. On the internet, there is a suite of the protocols known as TCP/IP protocols that are consisting of transmission control protocol, internet protocol, file transfer protocol, dynamic host configuration protocol, Border gateway protocol and a number of other protocols. In the telecommunication, a protocol is set of rules for data representation, authentication, and error detection. The communication protocols in the computer networking are intended for the secure, fast and error free data delivery between two communication devices. Communication protocols follow certain rules for the transmission of the data.

Protocols Properties

There are some basic properties of most of the protocols.

•Detection of the physical (wired or wireless connection)
•Handshaking
•How to format a message.
•How to send and receive a message.
•Negotiation of the various connections
•Correction of the corrupted or improperly formatted messages.
•Termination of the session.

Following is the description of the some of the most commonly used protocols.

HTTP (Hyper Text Transfer Protocol)

Hypertext transfer protocol is a method of transmitting the information on the web. HTTP basically publishes and retrieves the HTTP pages on the World Wide Web. HTTP is a language that is used to communicate between the browser and web server. The information that is transferred using HTTP can be plain text, audio, video, images, and hypertext. HTTP is a request/response protocol between the client and server. Many proxies, tunnels, and gateways can be existing between the web browser (client) and server (web server). An HTTP client initializes a request by establishing a TCP connection to a particular port on the remote host (typically 80 or 8080). An HTTP server listens to that port and receives a request message from the client. Upon receiving the request, server sends back 200 OK messages, its own message, an error message or other message.

POP3 (Post Office Protocol)

In computing, e-mail clients such as (MS outlook, outlook express and thunderbird) use Post office Protocol to retrieve emails from the remote server over the TCP/IP connection. Nearly all the users of the Internet service providers use POP 3 in the email clients to retrieve the emails from the email servers. Most email applications use POP protocol.

SMTP (Simple Mail Transfer Protocol)

Simple Mail Transfer Protocol is a protocol that is used to send the email messages between the servers. Most email systems and email clients use the SMTP protocol to send messages to one server In configuring an email application, you need to configure POP, SMTP and IMAP protocols in your email software. SMTP is a simple, text based protocol and one or more recipient of the message is specified and then the message is transferred. SMTP connection is easily tested by the Telnet utility. SMTP uses the by default TCP port number 25

FTP (File Transfer Protocol)

FTP or file transfer protocol is used to transfer (upload/download) data from one computer to another over the internet or through or computer network. Typically, there are two computers are involved in the transferring the files a server and a client. The client computer that is running FTP client software such as Cuteftp and AceFTPetc initiates a connection with the remote computer (server). After successfully connected with the server, the client computer can perform a number of the operations like downloading the files, uploading, renaming and deleting the files, creating the new folders etc. Virtually operating system supports FTP protocols.

IP (Internet Protocol)

An Internet protocol (IP) is a unique address or identifier of each computer or communication devices on the network and internet. Any participating computer networking device such as routers, computers, printers, internet fax machines and switches may have their own unique IP address. Personal information about someone can be found by the IP address. Every domain on the internet must have a unique or shared IP address.

DHCP (Dynamic Host Configuration Protocol)

The DHCP or Dynamic Host Configuration Protocol is a set of rules used by a communication device such as router, computer or network adapter to allow the device to request and obtain and IP address from a server which has a list of the larger number of addresses. DHCP is a protocol that is used by the network computers to obtain the IP addresses and other settings such as gateway, DNS, subnet mask from the DHCP server. DHCP ensures that all the IP addresses are unique and the IP address management is done by the server and not by the human. The assignment of the IP addresses is expires after the predetermined period of time. DHCP works in four phases known as DORA such as Discover, Observe, Request and Authorize

IMAP (Internet Message Access Protocol)

The Internet Message Access Protocol known as IMAP is an application layer protocol that is used to access to access the emails on the remote servers. POP3 and IMAP are the two most commonly used email retrieval protocols. Most of the email clients such as outlook express, thunderbird and MS outlooks support POP3 and IMAP. The email messages are generally stored on the email server and the users generally retrieve these messages whether by the web browser or email clients. IMAP is generally used in the large networks.

IMAP allows users to access their messages instantly on their systems.

FDDI

Fiber distributed data interface (FDDI) provides a standard for data transmission in a local area network that can extend a range of 200 kilometers. FDDI local area network can support a large number of users and can cover a large geographical area. FDDI uses fiber optic as a standard communication medium. FDDI uses dual attached token ring topology. A FDDI network contains two token rings and the primary ring offers the capacity of 100 Mbits/s. FDDI is an ANSI standard network and it can support 500 stations in 2 kilometers.

UDP

The user datagram protocol is a most important protocol of the TCP/IP suite and is used to send the short messages known as datagram. Common network applications that uses UDP are DNS, online games, IPTV, TFTP and VOIP. UDP is very fast and light weight. UDP is an unreliable connectionless protocol that operates on the transport layer and it is sometimes called Universal Datagram Protocol.

TFTP

Trivial File Transfer Protocol (TFTP) is a very simple file transfer protocol with the very basic features of the FTP. TFTP can be implemented in a very small amount of memory. TFTP is useful for booting computers such as routers. TFTP is also used to transfer the files over the network. TFPT uses UDP and provides no security features.

SNMP

The simple network management protocol (SNMP) forms the TCP/IP suite. SNMP is used to manage the network attached devices of the complex network.

PPTP

The point to point tunneling protocol is used in the virtual private networks. PPP works by sending regular PPP session. PPTP is a method of implementing VPN networks.

R.Alagianambi
Pre final year

Wireless Sensor Network

A wireless sensor network (WSN) is a wireless network consisting of spatially distributed autonomous devices that use sensors to monitor physical or environmental conditions. These autonomous devices, or nodes, combine with routers and a gateway to create a typical WSN system. The distributed measurement nodes communicate wirelessly to a central gateway, which provides a connection to the wired world where you can collect, process, analyze, and present your measurement data. To extend distance and reliability in a wireless sensor network, you can use routers to gain an additional communication link between end nodes and the gateway.

National Instruments Wireless Sensor Networks offer reliable, low-power measurement nodes that operate for up to three years on 4 AA batteries and can be deployed for long-term, remote operation. The NI WSN protocol based on IEEE 802 and ZigBee technology provides a low-power communication standard that offers mesh routing capabilities to extend network distance and reliability. The wireless protocol you select for your network depends on your application requirements. To learn more about other wireless technologies for your application, read the “Selecting the

Right Wireless Technology" white paper.

WSN Applications

Embedded monitoring covers a large range of application areas, including those in which power or infrastructure limitations make a wired solution costly, challenging, or even impossible. You can position wireless sensor networks alongside wired systems to create a complete wired and wireless measurement and control system.

A WSN system is ideal for an application like environmental monitoring in which the requirements mandate a long-term deployed solution to acquire water, soil, or climate measurements. For utilities such as the electricity grid, streetlights, and water municipals, wireless sensors offer a lower-cost method for collecting system health data to reduce energy usage and better manage resources. In structural health monitoring, you can use wireless sensors to effectively monitor highways, bridges, and tunnels. You also can deploy these systems to continually monitor office buildings, hospitals, airports, factories, power plants, or production facilities.

WSN System Architecture

In a common WSN architecture, the measurement nodes are deployed to acquire measurements such as temperature, voltage, or even dissolved oxygen. The nodes are part of a wireless network administered by the gateway, which governs network aspects such as client authentication and data security. The gateway collects the measurement data from each node and sends it over a wired connection, typically Ethernet, to a host controller. There, software such as the NI Lab VIEW graphical development environment can perform advanced processing and analysis and present your data in a fashion that meets your needs.

A WSN measurement node contains several components including the radio, battery, microcontroller, analog circuit, and sensor interface. In battery-powered systems, you must make important trade-offs because higher data rates and more frequent radio use consume more power. Today, battery and power management technologies are constantly evolving due to extensive research.

Often in WSN applications, three years of battery life is a requirement, so many of the WSN systems today are based on ZigBee or IEEE 802.15.4 protocols due to their low-power consumption. The IEEE 802.15.4 protocol defines the Physical and Medium Access Control layers in the networking model, providing communication in the 868 to 915 MHz and 2.4 GHz ISM bands, and data rates up to 250 kb/s. ZigBee builds on the 802.15.4 layers to provide security, reliability through mesh networking topologies, and interoperability with other devices and standards. ZigBee also allows user-defined application objects, or profiles, which provide customization and flexibility within the protocol.

In addition to long-life requirements, you must consider the size, weight, and availability of batteries as well as international standards for shipping batteries. The low cost and wide availability of carbon zinc and alkaline batteries make them a common choice. Energy harvesting techniques are also becoming more prevalent in wireless sensor networks. With devices that use solar cells or collect heat from their environment, you can reduce or even eliminate the need for battery power.

Processor Trends

To extend battery life, a WSN node periodically wakes up to acquire and transmit data by powering on the radio and then powering it back off to conserve energy. The WSN radio must efficiently transmit a signal and allow the system to go back to sleep with minimal power use. Likewise, the processor must also be able to wake, power up, and return to sleep mode efficiently. Microprocessor technology trends for WSNs include reducing power consumption while maintaining or increasing processor speed. Much like your radio choice, the power consumption and processing speed trade-off is a key concern when selecting a processor for WSNs. This makes PowerPC and ARM-based architectures a difficult option for battery-powered devices. A more common architecture option includes the TI MSP430 MCU, which is designed for low-power operation. Depending on the specific processor, power consumption in sleep mode can range from 1 to 50 µW, while in on-mode the consumption can range from 8 to 500 mW.

Networking Topologies

The first, and most basic, is the star topology, in which each node maintains a single, direct communication path with the gateway. This topology is simple but restricts the overall distance that your network can achieve.

The mesh network topology remedies this issue by using redundant communication paths to increase system reliability. In a mesh network, nodes maintain multiple communication paths back to the gateway, so that if one router node goes down, the network automatically reroutes the data through a different path. The mesh topology, while very reliable, does suffer from an increase in network latency because data must make multiple hops before arriving at the gateway.

With the National Instruments WSN platform, you can customize and enhance a typical WSN architecture to create a complete wired and wireless measurement system for your application. NI software integration provides the flexibility to choose a Windows-based host controller for your WSN system or a real-time controller such as NI CompactRIO, giving you the ability to integrate reconfigurable I/O with your wireless measurements. With either host controller, you can use Lab VIEW and the NI-WSN software with Lab VIEW project integration and drag-and-drop programming to easily configure your WSN system, extract high-quality measurement data, perform analysis, and present your data.

In addition, Lab VIEW integration delivers the ability to stretch the connectivity of your WSN from the enterprise and database level all the way through the Internet to end client devices, such as an iPhone or laptop. You can use this complete system architecture to acquire data from virtually anywhere with an NI wireless sensor network, process and host that data on a server, and then access the data conveniently and remotely from a wireless smart device

Vignesh ,
Second Year.

NONTECHNICAL SCOOPS

When a knife disguised a pen!!

The quote goes like this "The child is the father of the man." said William Wordsworth. Parenting is the art by which the good character is inculcated to the child by its parent. But a recent tragedy in one of the private school in Chennai in which a student had guts to stab his teacher, made the whole country dumbstruck and raised issues regarding the mental behavior of students. Investigations revealed that the boy was inspired by the Hindi movie. It has steered his impulse away from him and he could murder his own teacher. Media has started a worse impact among the teenagers. They are meant for imparting education to masses, in lieu they had created a barbaric scenario. Let us not enjoy films of violence with children. When a teacher is mocked by group of students it becomes a comedy for society. Parents themselves feel agitated when a teacher corrects their wards. They too dishonor such teachers and their children in abusing them. Gone are days of matha, pitha, guru, deivam (mother, father, teacher, god).Society must always remember that teaching is only a profession which read various minds in a class and deals with different temperaments. We must stop blaming any school or a teacher for not imparting moral values. Most schools give importance to catechism, counseling, yoga and meditation. Talking about the media, which has created an adverse effect among students; it must be curtailed to stop the young generation from being surrounded by devil inside them.It is obnoxious to say 14 years old creeps upon his teacher and stabs her to death because she has written remarks on his diary. A deadly rage within his bosom grows day by day; he feels pleasure on him from home and school. This rage reaches breaking point; he gets a knife and stab the teacher only for doing her duty! Can we complain about the illness in education scenario, media or parents? Nevertheless, parents are backbone for creating a child's character but what went wrong which tempted a mere a kid to do a brutal task. Education is a possession which cannot be taken away from men. But their art has down many questioning about the integrity of society. All these acts caused blight on entire society. Questions that need answers if such endemic rage is stopped. Thus, in conclusion, we should join hands together to fight against these kinds of activities and protect our youth from being rottener. But dedicated and determined teachers will continue to render their service to student community without bothering about brickbats, humiliation, threats and even killing.

-Magazine team

The Moral Values of Engineers Ruined!!!

Engineering is the art of creativity and innovation. But recently the news of robbery by engineering graduates made a shocking and initiated a questioning about professionalism and moral ethical values. The moral values are inculcated to students' right from their schooling days. But what makes these students that a devil spirit of thief and robbery enters them. To analyze with, many say unemployment causes them to steer away the moral values and struck up into hallows of unorthodox arena. The reasons trace that many pass outs being unemployed chose a way of robbery, murder and activities of chain snatching etc. Immediate measures must be taken to halt these activities. Student-turn-robbers on getting caught approve their perpetual state of unemployment, poverty. But what about the nation's self respect which is dumped into the soil? Government must provide opportunities to these young budding engineers. What about Dr. Kalam dream of becoming a super power in 2020! If these activities persist, we would be marching towards the pathetic country in 2020. Another injustice that must be thrown a light is brain washing of terrorists. Some of anti-elements of society knowing about the present conditions train the youngsters wrongly in correct way. The young ones without understanding the future consequences harm the nation which disrupts the national harmony of country. So an overall development of country is at stake, a stagnant view and devastation of youngsters. How a nation can progress if these conditions persist. It's time to join hands to perish these unhealthy activities; otherwise it would be high time we lose our integrity and dignity amongst the world scenes. Being technocrats and a future leaders, young buds should not the dejection of the nation. So on a concluding node, the development of nation depends upon the perseverance and determination of youth in healthy way, not in contrast to what the present time exist.

-***Magazine team***

Kudankulam Nuclear Power Project

Energy in present era:

In the present era of improving technologies, one needs to find better sources to harness more energy. There are many generation stations which are meant for generating bulk electrical energy to be applied at different loads.

The types of generating stations in India are:-

a) Thermal Power station
b) Hydro Power station
c) Wind Power station
d) Nuclear Power station

Plus and Minus:

The country's coal resources are estimated as about 112 billion tons. Assuming 50% recovery of the reserves in ground, the coal resources may last about 125 years. Though electricity generation based on natural gas is now considered on a limited scale due to its contemporary availability, this can't be a long term solution. In the case of oil, its use is now limited to the transportation sector and petrochemical industries. As far as Hydel power is concerned, most of the untapped Hydel potential is situated in relatively inaccessible locations in the extreme northern and eastern parts of the country. Also, one has to take into account the concerns associated with taking up new hydel scheme. Non conventional energy resources like solar, wind, tidal biogas and biomass have their own limitation and their contribution can at the best be marginal.

Nuclear Energy:

In case of nuclear energy, where the principle of nuclear fission is used for electricity generation, the electricity generated is exponentially greater than the input. There is negligible waste generation, which is reused or safely disposed. Thus, the electricity generated is more stable, eco-friendly and reliable. India started its nuclear venture in the late 1960s, starting from Tarapur, Maharashtra.

KKNPP:

Kudankulam Nuclear Power Project is a joint venture for construction of nuclear power stations consisting of two units of 1000MW Pressurized Water Reactor (VVER) at Kudankulam, in Tamil Nadu.

Kudankulam & its safety:

Any plant site, before construction, needs to undergo a number of tests and analysis, both about the technical and topographical aspects. In that way Kudankulam site was found suitable for the following reasons:-

a) Availability of hard-rock at reasonable depth, providing good foundation conditions.
b) Low seismic potential i.e. the site is situated in seismic level 2, which has least probability of event of earthquake.
c) Not subjected to severe cyclone, storms, tidal waves etc.
d) Economical lengths for the intake structures for drawing of condenser cooling water from the sea.
e) No population within 2km from the plant site.
f) Plant site on barren land with no agricultural produce.
g) Good accessibility by road, rail and sea.
h) No danger to safety of the nuclear plant from man induced events such as air impact.

VVER1000 Reactor:

The VVER1000 reactor belongs to the family of pressurized water reactors , which is the predominant type in operation all over the world. It has been christened in the Russian language where V indicates Volga, which means ‘water’ in the Russian language. Thus the reactor has water as both moderator and coolant.

Agitators:

The KKNP site is surrounded by number of small fishing hamlets. After the Fukushima-Daiichi nuclear accident and inadequate awareness, the people there got frightened and humpty-number of fear and insecurity crept into them. Their fears were,

- Depletion in fish-breeding count.
- Fear of health disorders due to radiation released.
- Fear of an aftermath of an accident, if occurred in KKNPP.
- Fear of displacement and livelihood.

These fears formed into an agitating group whose sole objective is to shut down the nuclear reactor once and for all.

Awareness:-

The current agitation is as a result of lack of awareness about nuclear energy and the way it is being handled. In order to solve the problem, the Indian govt. took many measures and formed many committee, all of which tagged KKNPP to be the safest and eco friendly nuclear reactor. Dr. A.P.J. Abdul Kalam, the Former President of India and a renowned scientist, who supports renewable and eco friendly energy, on his recent visit to KKNPP, inspected the project safety and emergency technologies and has inferred the following.

1. KKNPP will not be affected by earthquake as it is in zone 2 (least vulnerable zone). Nellaiyappar temple, a 1000 year old structure, is standing aloft in the same zone and is unfettered.

2. No nuclear waste is being dumped into the sea. The nuclear by-products and wastes are being reprocessed and reused and the ultimate waste is being crystallized to be buried several meters below the earth’s surface with proper containment.

3. The sea water is being used only to cod down the coolant water and is not coming under the direct contact of the radiation. Thus, the sea water is radiation free.

4. The hot sea water, after processing is not directly allowed into the sea. The temperature is reduced to normal temperature and at the time of release into the sea, the temperature variation is maintained to be less than 7 degree Celsius, which helps enhance fish breeding. This is a boon to the fisher men. KKNPP is the safest nuclear power plant. The new safety systems: Passive Heat Removal System and core catcher are the finest of the technology.

India is a developing nation and for a developing nation the energy demand is always on a peak. This energy need must be satisfied through an eco-friendly safe and cost effective manner. "Well! We take an accident in Japan in the year 2011, in a 40 year old reactor at Fukushima, arising out of extreme natural stresses, to derail our dreams to be an economically developed nation? The power of the nuclear reactor is mighty and future of humanity lies in harnessing it in a safe and efficient manner."

-Magazine team

LIGHT FROM DREAM

One fine night Vivek, an engineering student was keenly attached in surfing some topics via internet. He was viewing about a mammal "ELECTRIC EEL" that resides on beneath layer of the sea. It's a special type of mammal that usually catches its prey by producing an electric shock from its body. When vivek was busy in surfing, his father advised him to study as his exams were nearing. When he was about to study, the power went off. With a great sound of sigh he said "what a life is this, when I was ought to study power went. I don't know how tenth and twelveth students are going to prepare for their final exams". He had no other go, so went to bed and slept. Suddenly his father woke him up saying current came. He woke from sleep and took his book to study. But his mind was not here. He started to think a solution for this current problem.

Next day, he started his research with "generating electricity from shock produced by electric eel". His idea was when eel produces a shock; it stretches for about 50m in circumferential diameter.

The ions present in shock will be diffused in salt water of the seas. We know salt water is good conductor of electricity. So Vivekused this water to generate current and his research marched towards success. Soon, he presented a seminar for his invention. He explained his project to the co-students and addressed them as

"*Friends, I have did nothing. My role in this power cut problem is very small. But our country constructed a wonderful nuclear plant at Kudankulam, only for welfare of people. But our people not understanding the current situation and blaming for safety precautions. The invent which I had made is more money consuming. So, it's in our hands to make understand the people the real fact*".

His principal appreciated him and vivek was overwhelmed with joy. Suddenly he fell down from his bed and all things went blank. At last, he understood everything was a dream. Vivek thought "I will make my dream true one day and it's in the hands of youth to make our country cherished "and continued his sleep.

-Magazine team

The Mullaperiyar Controversy

The History and Construction of Mullaperiyar Dam

Mullaperiyar Dam or Mullai Periyar Dam a masonry gravity dam located on the banks of Periyar River in the Kerala state of India. It is located 881 m (2,890 ft) above mean sea level on the Cardamom Hills of the Western Ghats in Thekkady, Idukki District of Kerala, South India.

It was constructed between 1887 and 1895 by the British Government to divert water eastwards to Madras Presidency area (the present-day Tamil Nadu). It has a height of 53.6 m (176 ft) from the foundation and length of 365.7 m (1,200 ft). The Periyar National Park in Thekkady is located around the dam's reservoir. The dam is located in Kerala on the river Periyar, but the dam is controlled and operated under a period lease by neighboring Tamil Nadu state.The Periyar river which flows westward into the Arabian sea was diverted eastwards to flow towards the Bay of Bengal to provide water to the arid rain shadow region of Madurai in Madras Presidency which was in dire need of a greater supply than the small Vaigai River could give. The dam created the Periyar Thekkady reservoir, from which water was diverted eastwards to via a tunnel to augment the small flow of the Vaigai River. Currently, the water from the Periyar (Thekkady) Lake created by the dam, is diverted through the water shed cutting and a subterranean tunnel to Forebay Dam near Kumili (Errachipalam) in Tamil Nadu. From the Forebay dam, hydel pipe lines carry the water to the Periyar Power Station in Lower Periyar, Tamil Nadu. This is used for power generation (175 MW capacity) in the Periyar Power Station. From the Periyar Power Station, the water is let out into Vairavanar river and then to Suruliyar and from Suruliyar to Vaigai Dam.

Safety Concerns:

After the 1979 Morvi Dam failure which killed up to 15,000 people, safety concerns of the aging Mullaperiyar dam's and alleged leaks and cracks in the structure were raised by the Kerala Government. A Kerala government institution, Centre for Earth Science Studies (CESS), Tiruvananthapuram, had reported that the structure would not withstand an earthquake above magnitude 6 on the Richter scale. The dam was also inspected by the Chairman, CWC (Central Water Commission). On the orders of the CWC, the Tamil Nadu government lowered the storage level from 142.2 feet to 136 feet, conducted safety repairs and strengthened the dam.

Strengthening measures adopted by Tamil Nadu PWD from 1979 onwards include cable anchoring of the dam's structure and RCC backing for the front slope. During a recent scanning of the Mullaperiyar dam using a remotely operated vehicle by the Central Soil and Materials Research Station on directions from the Empowered Committee of the Supreme Court, the Kerala Government observer opined that "mistakes in the strengthening works carried out by Tamil Nadu" in 1979 damaged the masonry of the dam.

Current safety concerns hinge around several issues. Since the dam was constructed using stone rubble masonry with lime mortar grouting following prevailing 19th century construction techniques that have now become archaic, seepage and leaks from the dam have caused concern. Moreover, the dam is situated in a seismically active zone. An earthquake measuring 4.5 on the Richter scale occurred on June 7, 1988 with maximum damage in Nedumkandam and Kallar (within 20 km of the dam). Consequently several earthquake tremors have occurred in the area in recent times. These could be reservoir-induced seismicity, requiring further studies according to experts. A 2009 report by IIT Roorkee stated that the dam "was likely to face damage if an earthquake of the magnitude of 6.5 on the Richter scale struck its vicinity when the water level is at 136 feet".

The Interstate Dispute:

Tamil Nadu has insisted on raising the water level in the dam to 142 feet, pointing out crop failures. One estimate states that "the crop losses to Tamil Nadu, because of the reduction in the height of the dam, between 1980 and 2005 are a whopping ₹40,000 crores. In the process the farmers of the erstwhile rain shadow areas in Tamil Nadu who had started a thrice yearly cropping pattern had to go back to the bi-annual cropping."

In 2006, the Supreme Court of India by its decision by a three member division bench allowed for the storage level to be raised to 142 feet (43 m) pending completion of the proposed strengthening measures, provision of other additional vents and implementation of other suggestions. The Kerala Government states that it does not object to giving water to Tamil Nadu, their main cause of objection being the dam's safety as it is 116 years old. Increasing the level would add more pressure to be handled by already leaking dam. Tamil Nadu wants the 2006 order of Supreme Court be implemented so as to increase the water level to 142 feet (43 m).

On 18 February 2010, the Supreme Court decided to constitute a five-member empowered committee to study all the issues of Mullaperiyar Dam and seek a report from it within six months. The Bench in its draft order said Tamil Nadu and Kerala would have the option to nominate a member each, who could be either a retired judge or a technical expert. The five-member committee will be headed by former Chief Justice of India A. S. Anand to go into all issues relating to the dam's safety and the storage level.

In pursuance of Kerala's dam safety law declaring Mullaperiyar dam as an endangered dam, in September 2009, the Ministry of Environment and Forests of Government of India granted environmental clearance to Kerala for conducting survey for new dam downstream. Tamil Nadu approached Supreme Court for a stay order against the clearance; however, the plea was rejected. Consequently, the survey was started in October, 2009. On Sept. 9, 2009 stated it had already communicated to the Government of India as well as to the Government of Kerala that there is no need for construction of a new dam by the Kerala Government, as the existing dam after it is strengthened, functions like a new dam.

Conclusion:

Kerala has been saying that Dam's structure is extremely week and it can break any time which can causes severe catastrophe in the region and Tamilnadu's view is that it can be maintained by adequate repair. As a part of the society and people we believe that both government can handle the situation well and resolve it soon so that the tension between states is reduced without any damage to anybody.

"Can't I just email you a link to my blog, Miss?"

Current Events:

2200-Year-Old Tamil-Brahmi Inscription found at Samanamalai in Madurai District:

A 2200-year-old Tamil-Brahmi inscription belonging to the second century BC, was discovered at Samanamalai in Madurai district. The inscription is associated with Jainism as it was found at Samanamalai ("Jaina Hill"), 15 km from Madurai.

IUCN Report: 14 Species of Birds from India on the Verge of Extinction:

IUCN (International Union for Conservation of Nature) on 10 November 2011 released the red List of threatened species, which includes 14 species of birds from India. These birds are critically endangered. The endangered birds include, Indian Vulture, Red-headed Vulture, Pink-headed Duck, Sociable Lapwing, Jerdon's Courser, Siberian Crane, White-belled Heron, Christmas Frigatebird, Bengal Florican, Forest Owlet, Spoon-billed Sandpiper, Himalayan Quail, White-rumpedVulture.The major threats faced by these bird species include poaching, habitual destruction, indiscriminate use of pesticides and chemicals harmful for birds. However, the national bird peacock is not under the threat of extinction.

Number of People living Below Poverty Line in Sikkim came down to 80000:

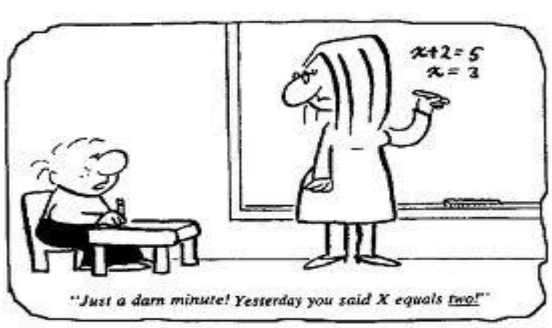
As per the poverty estimates released by the Planning Commission for year 2009-10, the number of people living below poverty line in North-Eastern State Sikkim, has come down. The figure has come down with 80000 people now living below poverty line.

M.C. Mary Kom and Sarita won Gold Medals in the Asian Women's Boxing Championship

Indian women boxers M.C. Mary Kom and Sarita on 25 March 2012, clinched two gold medals in the Asian women's boxing championship at Mongolian capital Ulaanbator.MaryKom, the five-time world champion, routed the two-time world title winner and 2010 Asian Games gold medalistRen Cancan of China 14-8. Sarita, the 2006 world title winner, smashed defending champion ChorievaMavzu of Tajikistan 26-15.

Ranbaxy emerged Fastest Growing Pharmaceutical Company in India in First Quarter of 2011:

According to market research firm IMS, Ranbaxy Laboratories emerged the fastest growing pharmaceutical company in India in the first quarter of 2011. According to market research, Ranbaxy clocked sales growth of 17.3% from January to March in 2011 over the same period last year, although in terms of market share (4.78%) it remained second.



India's GDP Growth to make the Indian Banking Industry Third Largest in the World by 2025:

A study titled Being five star in productivity — road map for excellence in Indian banking was released FICCI-IBA-BCG on 22 August 2011, the eve of IBA-FICCI annual banking conference. The theme for the banking conference was decided to be Productivity Excellence. According to the study, India's gross domestic product (GDP) growth will make the Indian banking industry third largest in the world by 2025. The report chalked out an action agenda for banks, based on insights from an extensive productivity benchmarking exercise conducted across 40 banks.

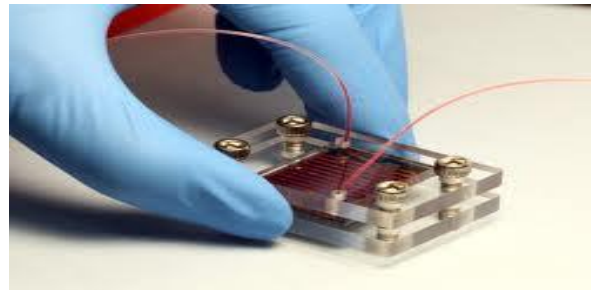
Coal India- Fifth Most Valuable Indian Company:

Coal India that made its debut on stock exchanges on 4 November 2010 became the country's fifth most valuable company with a market capitalisation of Rs 2.03 lakh crore within the first hour of trade in the opening day. State-run Coal India left behind Infosys, power utility NTPC Ltd, the country's largest private lender ICICI Bank, FMCG major ITC and engineering giant L&T to enter into the club of top 10 most valued firms. ONGC Ltd and State Bank of India were the only two public sector entities that are ahead of Coal India in terms of full market capitalisation.

A Gas-filled Aspirin was invented by the Scientist to fight Cancer:

A scientist from Iran developed a gas-filled aspirin that can enhance the cancer-fighting ability of the drug. The new Aspirin was dubbed NOSH, which stands for Nitric Oxide and Hydrogen Sulphide. The new Aspirin reduces the harmful side effects of taking aspirin. Aspirin is effective against Cancer. But it can also cause bleeding in the gut and ulcers. The gas-filled Aspirin protects the gut from damage by producing nitric oxide and hydrogen sulphide. NOSH-aspirin was 100000 times more effective than the original drug in case of colon cancer.

Researchers invented New Technique to run Ultra-fast Internet:



Researchers in the second week of March 2012 invented fibre-optic technology, under a project called Sardana to run ultra-fast Internet. It demonstrated speed of upto 10 Gigabits per second (Gbps), around 2000 times faster than today's fastest Internet speed. The research showed that by using existing fibre infrastructure Internet's speed could be increased. Within next three years, yearly global internet traffic will be measured in Zetta bytes(one trillion Gigabytes), which is a four-fold increase from today. You Tube and Netflix will have most of the traffic.

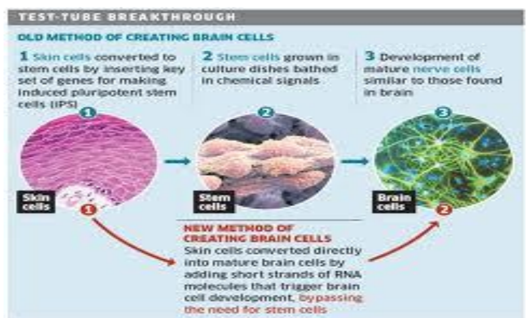
DRDO Conducted Test of Interceptor Missile Successfully:

DRDO on 10^{th} February 2012 conducted a successful test launch of the interceptor missile. DRDO's Air Defence Missile AAD-05 successfully hit the ballistic missile and destroyed it at a height of 15 kms off the Coast of Orissa near the Wheelers Island.A modified Prithvi missile mimicking the ballistic missile was launched at 1010 hrs from ITR Chandipur. Radars located at different locations tracked the incoming ballistic missile. Interceptor missile was ready to take-off at Wheelers Island. Guidance computers continuously computed the trajectory of the ballistic missile and launched AAD-05 Interceptor Missile at a precisely calculated time. With the target trajectory continuously updated by the radar, the onboard guidance computer guided the AAD-05 Interceptor Missile towards the target missile. The onboard radio frequency seeker identified the target missile, guided the AAD-05 Interceptor Missile close to the target missile, hit the target missile directly and destroyed it. Warhead also exploded and destroyed the target missile into pieces. India is the fifth nation to have these ballistic missile defence capabilities in the world. Radar and Electro Optic Tracking Systems (EOTS) tracked the missile and also recorded the fragments of target missile falling into the Bay of Bengal. It is one of the finest missions where the interceptor has hit the incoming ballistic missile directly and destroyed it at a 15 kms altitude. The mission was carried out in the final deliverable user configuration mode.

UN Security Council supported Kofi Annan's Plan to end Syria Unrest:

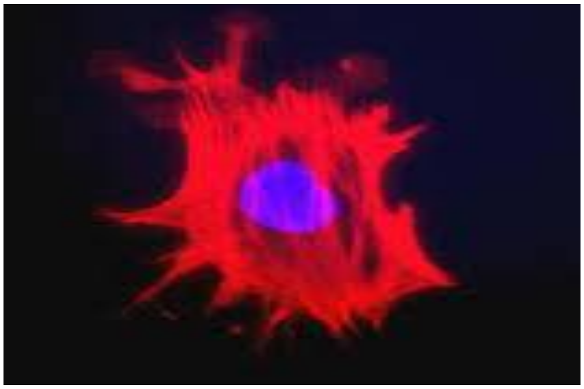
The United Nations Security Council on 20 March 2012 adopted a statement supporting U.N.-Arab League envoy Kofi Annan's plan to try to end Syria's year-long unrest. The statement says Syria will face further sanctions if it rejects Annan's six-point peace proposal. The plan calls for a cease-fire, political dialogue between the government and opposition, and access for humanitarian aid agencies. Kofi Annan sent a five-member team of international experts to Syria on 18 March 2012 to try to secure a cease-fire between government and rebel forces.

Scientists transformed Skin Cells directly into Brain Cells :



Scientists recently transformed skin cells directly into brain cells. This scientific finding could bypass the need for stem cells. Stem cells can be turned into any other specialist type of cell ranging from brain to bone. Therefore, these cells are used in a range of treatments.

Scientists found Cells that can help prevent Spread of Cancer :



A team of scientists in the second week of January 2012 identified a group of cancer cells that play a main role in preventing the development of the disease. This scientific finding could lead to re-evaluation of common cancer treatment for patients. It pointed out that anti-angiogenic therapies that target those tumour cells- called pericytes- inadvertently make tumours more aggressive. The study was published in the journal Cancer Cell. It underscores the need of more research on tumour-cell composition to bring out more effective therapies.

UNHRC resolution against Sri Lanka's Human rights was adopted in Geneva:

UN human rights council in Geneva adopted a resolution against Sri Lanka's Human rights record. 24 countries including India voted for the resolution and 15 against. 9 countries abstained from voting. The resolution is about Human Rights violations committed by the Sri Lankan Army in its war with LTTE, which ended in May, 2009.However, India also emphasized that resolutions of this nature should fully respect the sovereign rights of states and contribute to Sri Lanka's own efforts in this regard. India's voting in favour of US Resolution on Sri Lanka in Geneva was in line with its stand.

India and the World Bank signed an IDA Credit of 152 Million US Dollars

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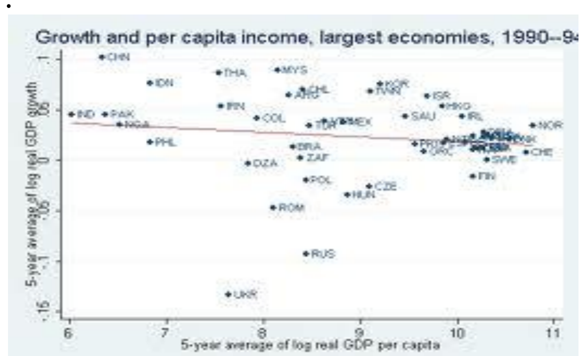


India and the World Bank on 21 March 2012 signed an IDA credit of 152 million US dollars to finance the Indian government's efforts to help improve the efficiency, quality and accountability of health services in Uttar Pradesh. The Project will include two main components. First, it will focus on improving the Department of Health`s management and accountability systems. The project will seek to improve, planning capacities, information flows and data inputs to enable policy-makers to improve monitor and manage service delivery in UP.

-Magazine team

IS INDIA A SOFT NATION

Is India a soft Nation? A hard question isn't it? But here is the answer; it is not a soft Nation! India is not as it looks like, just only the land looks united without any disputes. Without Indians being soft the nation won't be soft. Our country doesn't tackle war and terror with soft hands but it is pretending to be soft by hiding its hard master minds, Cargill, hoops a great war but even though some sacrificed their lives for the nation there was a hard minds stealing people money in name of them, Let it be on one side, coming to internal disputes.

India is very staunch in its caste, creed, racial, regional system and the toughest in separating the poor from the rich. Every day there are many who are dead in name of religion, caste disputes, and interstate disputes. A poor who works a day long and takes the quarter of his wage doesn't think India is a soft Nation. A family who has lost one of their beloved ones because of caste, religion, language won't think India is a soft Nation. A citizen who has been suppressed because of his caste and creed even though he has talents does not think India as a soft Nation. YES, India is a soft nation but to those who are rich. At last I would end by quoting that

"When the going gets tough; the Tough gets going."

Adwiti mehta
Pre final year

DICKER FROM REGIME

Have you ever wondered why when? Petrol prices go up or down they doso uniformly across the retail outlets of the three oil marketing companies Indian Oil, Hindustan Petroleum and Bharat Petroleum? If they are three a different companies with their own refineries, distribution systems, then surely their cost and selling price must be different? Oil companies have to be fixing their rate based on parameter. But unfortunately that there is no such parameter."We are in position to buy it with no alternatives".Think price of petrol in Bangladesh in 52.45 and in India its goes to 70....And it will increase today and goes on.

What is the reason behind? Currently domestic price are determine by price swings in the Singapore oil market. This is not in any other industry. Let take the case "steel". The price of finish steel produced by an Indian company will be determined by its cost of raw materials such as iron ore and the coal, cost of power used to produce steel plus other items of cost such as the Salaries and other factor.

Do you think why petrol rate changes in India?

The oil companies simply take the price of petrol in Singapore market, apply rupee-dollar exchange rates to that price, add other costs such freight and import duties, there comes the price that they should charge the domestic consumers.

Why should we in India pay for petrol?

A price that has no resemblance to either the cost structure of Indian Oil (or the Hindustan Petroleum Bharat Petroleum) or to competition in the market? Price in Singapore dance to various tunes ranges from fundamental reasons such as the consumption levels by countries in the region or shutdown of refineries or the pipelines to market-related reasons such as levels of speculation and money flows into the commodities market. How are we in India affected by these when we do not import a litre of petrol or diesel from Singapore? If this condition persists our petrol hike would reach Rs.84 next year.

"EVERYONE CAN RISE A QUESTION AND EVERY QUESTION SOMEONE WILL HAVE THE ANSWER AND THE SOMEONE IS ENGINEER".

S.Venkateshprasad
Pre final year

POEM OF PATRIOTISM

March ahead my noble soldier,
As our safety rests on your shoulder.
You live by chance and love by choice
Where there is nothing to rejoice.
You have the heart to bear all pain
Moving through sun, snow rain
Yours is a breed that fights like a king.
For you, it's never a personal thing
You are my role model without a question
As you lead, protect and kill by profession
Be it a regular duty or a cl operation.
With death always looming near,
You keep vigil without fear
You are the one who lives for the country's name.
Not caring for money, glamour or fame
You are the one-the leader of the game
And I would like to become the same.

S.SivaSethuRamalingam
Pre final year

FOR YOU MY DAD'

I was crawling,
I needed a lift,
You just gave it!
I was struggling,
I needed a push,
You just did it!
I was walking,
I wanted to run,
You just showed me how!
I was down and out
I wanted some help
God! You were there again
I was up in the stage,
I wanted a gift,
Your hug was nothing less!
I am almost done,
I wanted only you,
God! Where are you???

S.Karthikeyan
Pre final year

'ME'

In I walk,
With a lot to talk
But in my work,
Nothing upto the mark
Waiting for someone
Who can make me 'No.1'
Only showed me no one
It's then it struct in me
That it was only me
Who could get me,
Where I wanted to be…

S.Kartikeyan
Pre Final Year

BEST OF 2011

Best book of 2011:
The art of fielding by Chad Harbach.Little, Brown & Company.

Best innovative technology:
Motorola Atrix 4G.

Best invention of 2011:
The Stark Hand, Created by Mark Stark.

Best Discovery of 2011:
Area 51 Plane Crash Revealed.

Best Mobile:
Samsung Galaxy Note

Best Sold Software of 2011:
TurboTax Home & Business Federal + E-file + State 2011.

Best country to live:
China.

Best commodity:
Gold and Silver.

Best cricket team for test:
England.

Best cricket team for ODI:
Australia.

Best football national team:
Spain.

Best football team:
Bayern Munich

Courtesy : Google News
-Magazine team

கதைகேட்டுப்பழகியவர்கள்...!!!

பொதுவாகவே கதைகளை
இருவகைப்படுத்தலாம்..
ஒன்று,
தனக்கு இழைக்கப்பட்ட அநீதியை
தட்டி கேட்க வலுவில்லாத
தனியொருமனிதனின், சோகக்கதை...

மாறாக, மற்றொன்று,
இளித்தவாய் சாமானியனை
எளிமையாக ஏமாற்றியவனின்
ஏகபோக வெற்றிக்கதை...

கதைகள் மாறுபடும்,
கதைசொல்லி மாறுவான்...
ஆனால்,
ஊங்கொட்டி கதை கேட்பது
என்றும் நாம் தான்... இல்லையா???

ஏனெனில், "கேட்பதுஎளிது..!!"

ஏமாந்தவனின் சோகக்கதைகளுக்கு,

ஆரம்பத்தில் ஒரு "அச்சச்சோ"வும்..
இடையிடையே சில "அடப்பாவமே"வும்
இறுதியாக,
"கெட்ட கனவா நெனச்சு
மறந்திடுங்க"வும்

ஏமாற்றியவனின்
வெற்றிக்கதைகளுக்கு,
ஆரம்பத்தில் சில "ஆஹா
ஓஹோ"க்களும்
இடையிடையே சில
"அப்படிப்போடு"களும்
இறுதியாக,
(கொஞ்சம் அதிகம்
தான்..பரவாயில்லை...)
"உங்கள மிஞ்ச இந்த ஜில்லாவுல
ஆளில்லிங்க"வும்

கதைகேட்கும் நமது பங்களிப்புப்பாக
இருக்கக்கூடும்...

கதைசொல்லியை ஊக்கப்படுத்தும்
இந்தகுறிச்சொற்கள் வேண்டுமானால்
மாறலாம்...
மற்றபடி,
சாமானியனை நகையாடும்,
பலம் படைத்தவனின் பக்கமே
செல்லும்,
அநீதியை அதிகாரம் பண்ணச்சொல்லும்
அஹிம்சையை நிராகதிக்குத்தள்ளும்,
நம் இயல்பு,
என்றுமே மாறப்போவதில்லை....

நல்லது சொன்னால் தூங்குவது
நம்மவர் பழக்கம்...
தூங்கிவிட்டீர்களா???
அச்சச்சோ...
விழித்துக்கொள்ளுங்கள்...
பக்கத்தில் கதைகேட்க
ஆளில்லையாம்...!!!

S.Prasanna
Pre final year

वही है ,वही है ,वही है[That is]

हर बातों की बात वही है
दोस्तों का भी साथ वही है ,
सोचू जब अपने दिल से तो ,
बात वही है ,रात वही है |
दर्पण का अंधकार वही है ,
व्याकुलताकासार वही है ,
सोचू जब अपने आँखों सेतो ,
यार वही है ,प्यार वही है |
योद्धाओ का अहंकार वही है ,
सुरविरो का वार वही है ,
सोचू जब अपने मन से तो ,
दुनिया वही है, दिलदार वही है |
हर चाल का इंतज़ार वही है ,
हर काल कादीवार वही है.
सोचू जब
अपने साँसों से तो ,
क़र्ज़ वही है ,कर्ज़दार वही हैं ॥

Mukesh Kumar Aryan
Pre final year

दोस्ती [Friendship]

दिन हुआ है तो रात भी होगी,
होमत उदासक भी तो बात भी होगी,
इतने प्यार से दोस्ती की है खुदा की कसम
जिंदगी रही तो मुलाकात भी होगी ।
कोशिश की जिए हमें याद करने की
लम्हे तो अपने आप ही मिल जायेंगे
तमन्ना कीजिए हमें मिलने की
बहाने तो अपने आप ही मिल जायेंगे ।
महक दोस्ती की इश्क से कम नहीं होती
इश्क से ज़िन्दगी ख़तम नहीं होती
अगर साथ हो ज़िन्दगी में अच्छे दोस्त का
तो ज़िन्दगी जन्नत से कम नहीं होती
सितारों के बीच से चुराया है आपको
दिल से अपना दोस्त बनाया है आपको
इस दिल का ख्याल रखना
क्योंकि इस दिल के कोने में बसाया है
आपको अपनी ज़िन्दगी में मुझे शरिख
समझना
कोई गम आये तो करीब समझना
देदेंगे मुस्कराहट आंसुओं के बदले
मगर हजारों दोस्तो में अज़ीज़ समझना ..
हर दुआ का बुल नहीं होती ,
हर आरजू पूरी नहीं होती ,
जिन्हें आप जैसे दोस्त का साथ मिले ,
उनके लिए धड़कने भी जरुरी नहीं ॥

Sudhanshukumar
Pre final year

दिल की आवाज़[Voice of the heart]

इस शुभ काल की बेला में, मैं सबको लुभाना चाहता हूँ।
इतिहास बनाना चाहता हूँ, इतिहास बनाना चाहता हूँ।
उस सूर्य काल की बेला में , धरती पर मैं ने जब जन्म लिया,

तब सायंकाल की बेला में , माँथे पर मैंने जब कफन लिया,
अब रात्रीकाल की बेला में, इक दीया जलाना चाहता हूँ।
इतिहास बनाना चाहता हूँ, इतिहास बनाना चाहता हूँ।

दिल में मेरे जुनून सा था, साँसों में एक सुकून सा था,
मैं मंत्र ना लिए निकला था जब, तब सबने खूब सराहा था,
आगे आकर जब देखा तो ,बाहर कुछ और नजारा था।
इक गुरूदेव की आसथी तब, मुझको उसकी तालाश थी तब।

पर कहाँ गए वो गुरूदेव जो ज्ञान सिखाया करते थे।
वरदराज जैसे को भी विद्वान बनाया करते थे।
न्यूटन आइंसटीन के उस युग को फिर से दुहराना चाहता हूँ।
इतिहास बनाना चाहता हूँ, इतिहास बनाना चाहता हूँ।

Saurabh kumar
Pre Final Year

చిగురించినస్నేహం[*Friendship]*

కిరణానికిచీకటిలేదు...
సిరిమువ్వకిమౌనంలేదు.....
చిరునవ్వుకిమరణంలేదు.......
మన "స్నేహానికిఅంతంలేదు...
మరిచేస్నేహంచేయకు,
చేసేస్నేహంమరవకు
.విరబూసినవెన్నెలకరిగిపోతుంది....
వికసించినపుష్పంవాడిపోతుంది... ..
కానీ...."చిగురించినస్నేహం"
చిరకాలంనిలిచిపోతుంది......

అలలప్రయాణంతీరంవరకే,
మెరుపుప్రయాణంమెరిసేవరకే,
మేఘప్రయాణంకురిసేవరకే,
కలలప్రయాణంమెలకువవరకే,
ప్రేమప్రయాణంపెళ్ళివరకే,
కానీస్నేహప్రయాణంమరణాంతరంవరకు...

V.P.Kalyankumar
Pre final year

.మనసులోనిభావాలెన్నో

[Feelings of mind]

మనసులోనిభావాలెన్నో
మరువలేనిగాయాలెన్నో
వీడలేనిసేస్తాలెన్నో
వీడిపోనిబంధాలెన్నో
మరపురానిపాటలెన్నో

మధురమయినక్షణాలెన్నో

కవ్వించేకబుర్లెన్నో

మాయమయ్యేమార్పులెన్నో

అవసరానికిఆడినఅబద్ధాలెన్నో

ఆస్చర్యపరిచేఅద్భుతాలెన్నో

మాటల్లోచెప్పలేనిముచ్చట్లెన్నో

ముసుగువేసినమనసుకుమరువరానిజ్ఞా

పకాలెన్నో ఎన్నో ఎన్నెన్నో ఇంకెన్నో...

మనిషిజీవితంలోమరువలేనిఇంకెన్నో

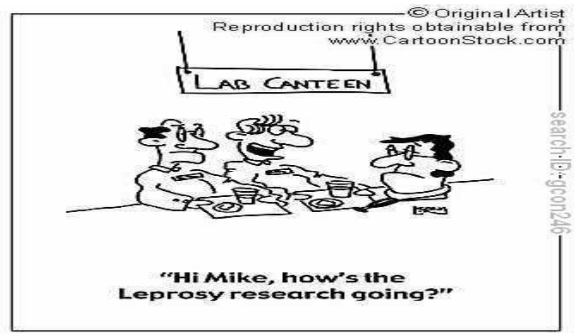
ఇదేజీవితం..దీనినిఅనుభవించుఅనుక్షణం

V.P.Kalyankumar
Pre final year

மனிதா!

சிறகடிக்கும் சிட்டுக்குருவி
சிறகொடிந்து கீழே விழும்போது
சின்னஞ்சிறு மழலையும் கண்கலங்கும்
சிந்தியுங்கள்!
மாந்தரென மண்ணில் பிறந்து,வாழ்ந்து
மாண்டு விடுவதில் என்ன அர்த்தம்!!-அந்த
மழலையின் மாசற்ற மனதை பார்
மனிதநேயம் பாராட்டு
மலர்ந்த முகத்துடன் மற்றவருக்கு உதவு
அதுவே அர்த்தமுள்ள வாழ்வு.......

த.விஜயபாரதி,
விரிவுரையாளர்



சுதந்திரம்

சுதந்திரமாய் வாழ ஆசைப்பட்டேன்
என் தாய் தந்தையைப் பிரிந்து
ஆனால் தவறு செய்துவிட்டேன்
சுதந்திரத்தை தவறாகப் புரிந்துகொண்டு...

S.கார்த்திகேயன்,
மூன்றாம் ஆண்டு

கருப்புத் தட்டு

நினைவுக்குவரும்,
நாம் முதலில் படித்த எழுத்துக்கள்.....
நினைவுக்குவரலாம்,
நாம் முதலில் பயன்படுத்திய பேனா.....
நினைவிருக்கிறதா???
நாம் முதலில் எழுதத் பயன்படுத்திய
அந்த சிறிய கருப்பு தட்டு......!!

S.கார்த்திகேயன்,
மூன்றாம் ஆண்டு

பதிலா???

முடியாது....
நடக்காது....
கஷ்டம்....
நஷ்டம்....

கவலை….
இல்லை….
பயம்….
துன்பம்….

இது என்னுடைய கேள்விகளுக்கு பதிலா???
இல்லை என்னை கேள்வி குறியாக்கும் மாய வலையா???

S.கார்த்திகேயன்
மூன்றாம் ஆண்டு

என் நண்பன்

உலகில் எனக்கு ஒரு உலகமாய் வந்தவனே
என்னை எனக்கு புரிய வைத்தவனே
என் திறமையை வெளி கொண்டு வந்த ஆசானே
என் வலிக்குமருந்து போட்ட மருத்துவனே
பாசம் காட்டுவதால் எனக்கு நீ தாயாகிறாய்
நல்வழியில் நடத்துவதால் எனக்கு நீ தந்தையாகிறாய்
என் வேதனையை பங்கெடுபதால் எனக்கு நீ பங்காளி
உன்னிடம் சண்டை போடும் பொழுது நீ என் மச்சான்
முறையோடு எதையும் கேட்பதால் நீ என் மாமா
என் உறவு அனைத்திலும் நீ ஒருவன்இருக்கிறாய்
உன்னை என் இதயத்தில் வைத்துள்ளேன்
நீ இருப்பதால் தோல்வி பயந்து நெருங்குவதில்லை
நீ என்றும் கலங்கரை விளக்குபோல் உயர்ந்துவிளங்க வேண்டும்
இன்றும் என்றும் உன் அன்புக்கு நான் அடிமை…

ஹரி சங்கர்,
இரண்டாம் ஆண்டு

"FORLORNED"

It eats my mind
it kills my thought
because when I want u
when I need u...

There are only memories
which I never thought
to be the last one
can't forget the last hug...

Perhaps you knew you are going away
but why couldn't he gave you little more time
so that I could have got one more hug
the last hug ... the last touch ..

You have stolen my mom
snatched my soul
how can I live on my own
missing my mom while being alone....
Was it me who cried....
was it me who tried....
travelled thousands of miles
to pray for her life....
still you didn’t blessed her
why it is irrevocable
why can’t I bring her back
and take a sound sleep in her lap ...

what the hell you want,
take everything whatever I have
whatever i would.... have ,
just give me my mother back......

I don’t know how
it’s going to be.....
Without you it’s like
a fruit without its tree.....

Mayank Sinha
PreFinal Year

CHAT BOX

Interview with Ms Deivanayagi, Final year placed with highest package in ATHENA HEALTH CARE.

1) **You are the only candidate recruited in Athena Health care. How do you feel?**

I am happy that I am recruited in Athena Health care. I would have been still happier if few more candidates were recruited.

2) **You are the candidate with highest package currently in university. What is your opinion on that?**

It's not the package which determines the ability of a person. Many of my friends who got placed in TCS and HCL and who were not allowed to appear for Athena health deserve such pay. So I am happy not because that I got such a wonderful package but I cleared wonderful interviewing process.

3) **What are your preparatory measures for Athena Health Care?**

To be frank my preparatory measures was almost null. Its the knowledge and experience that I gained over years that made me clear the rounds. I did not do any my international certifications till then (now I have completed one). But my friends did on **"Software Testing".** We (our friends and me) discussed the basics of Software testing which made me answer better in interview.

4) **What was going inside your mind when you cleared the preliminary levels and entering into HR interview?**

To say in detail, there was only one HR out of 9 recruiters. "All others are technical guys, he said" The HR was so friendly, he interacted very well. In the pre-placement talk itself, we were informed that we not appear well groomed for the interview.
All of these nullified my tension.

5) **Which is the key factor that made you to clear all the rounds?**

I have heard about many recruit processes and undergone one such process. From all those what I think "dedication and self-motivation" are the project and in programming lab. Throughout 4 years and the enthusiasm which always showed in solving a bug helped me clear the rounds.

6) **What was the level of difficulties in all the rounds?**

I think first round required only the ability to think. No specific preparations can be done. The second and third rounds required the ability to understand the program respectively. The compilers were given for the second round, so trial and error method was possible to solve the problem. The fourth and Fifth rounds were face to face interviews. It is the motivation, level, ability to work in team that was checked.

7) **What were the challenging moments in the recruitment process?**

Most challenging moments were my testing and programming round. Face to Face interviews were also challenging but time constraint in other rounds made them more challenging.

8) How did you face Athena health care after the two disappointments from TCS andHCL?

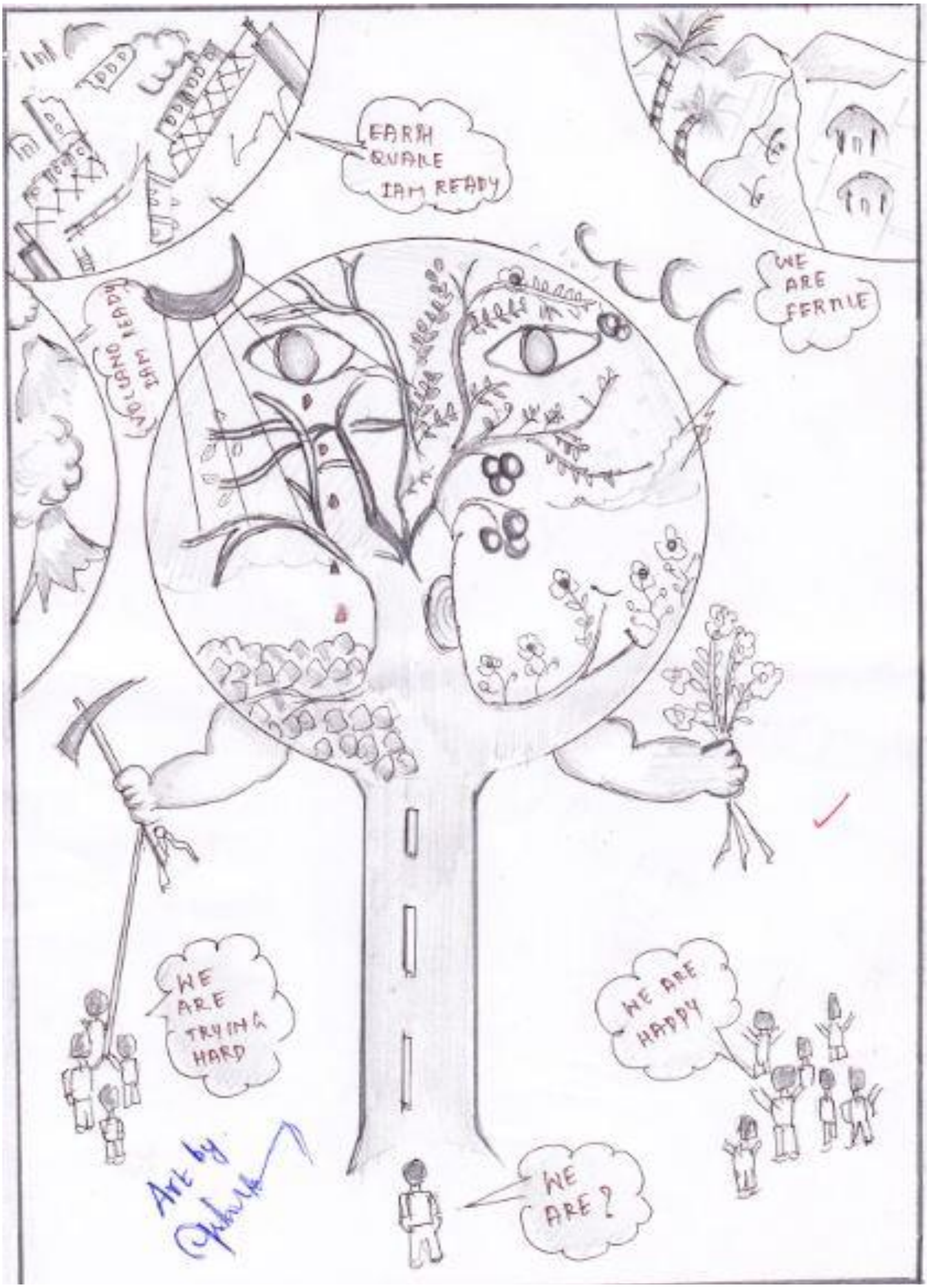
I was not at all disappointed by TCS because it was mass rejection. I was little disappointed by HCL as I didn't concentrate on campus recruitment. SoI faced Athena neither with expectations and nervousness.

9) What is your advice to your peers and juniors?

Enjoy learning! Do more and more projects. Do your final year project sincerely. The thirst for finding the solutions is very important. Quench your thirst with knowledge**..**

ART WORKS

Fathima sanofer
Pre final year



C.Ashok
Pre final year

M.Suganya @ Dharani
Pre Final Year

Ashish Tiwary
Pre Final Year

Balamurugan
Pre Final Year

<u>சிந்தனை துளிகள்!!</u>

1. எனக்கு நேரம் சரியில்லை
என்பது விரக்தியின் உச்சம்
நேரத்தை சரியாக நிர்வகித்தால்,
நேரம் உங்கள் எதிரியாகாது.

2. எது தெரியும் என்பதைவிட
எது தெரியாது என்பதை
அரிந்து கொள்வதே
ஆற்றல் தரும்.

3. எப்படி விழுந்தோம் என்பதை விட
ஏன் விழுந்தோம் என்பதை விட
எவ்வளவு வேகமாக எழுந்தோம்
என்பதுவே வெற்றிகான விதை.

4. மிகவும் அவசியம் இருந்தால் மட்டுமே
திறக்கும் பூட்டை உதட்டுக்குப் போடுங்கள்
நம் நீதிபதியான மனசாட்சியிடம்
அதன் சாவியை ஒப்படைத்து விடுங்கள்.

5. “பயம்” தேட வைக்கிறது
அறிவையும் அறியாமையையும்
உறவையும் நட்பையும்-ஏன்!
இறைவனையும் தேட வைக்கிறது.

6. நாம் உயர்ந்தால் யாருக்கு லாபம்
நாம் தாழ்ந்தால் யாருக்கு நட்டம்
சொல்வது யாராக இருந்தால் என்ன?
சொல்லப்படுவது நமக்காகத் தானே!

7. “பணம்” செலவைக் கற்றுத் தருகிறது.
“காலம்” திட்டமிடக் கற்றுத் தருகிறது.
“நூல்கள்” பண்பைக் கற்றுத் தருகிறது.
“மலர்கள்” அன்பைக் கற்றுத் தருகிறது.

8. “இன்று” உம் கையில் இருக்கும் போது
“நேற்று”க்காக வருந்தாமல்
“நாளை” வரும் காலங்களை
உனதாக்கிக் கொள்ளலாமே!

9. அன்பு என்ற ஒன்று
இல்லாது இருந்தால்
உலகம் என்பது
வெற்றிடமே!

10. கண்கள் ஊனமுற்று இருந்தாலும்
பார்வைகளில் தெளிவு பெற்றுள்ள
மாற்றுத் திறனாளியிடம் கற்றுக் கொள்வோம்
உலகைப் பார்க்கக் கற்றுக் கொள்வோம்.

11. வளையாமல் சென்றால்
அது நதி இல்லை
வலிக்காமல் இருந்தால்
அது வாழ்க்கை இல்லை.

12. விளை நிலங்களை எல்லாம்
வீட்டுமனைகள் ஆக்கி விட்டோமானால்,
நம் பேரக்குழந்தைகள்,தம் உணவை
எங்கே சென்று தேடப்போகின்றார்கள்?

13. நீரற்ற கேணி
இருப்பதல் பயன் என்ன?
அன்பற்ற மனிதன்
வாழ்வதால் பயன் என்ன?

14. குழந்தைகளே!
உங்கள் மகிழ்ச்சிக்குப் பின்னால்
உங்கள் பெற்றோரின்
கண்ணீர் இருக்கிறது.

15. நேர்மையானவர்கள்
மனசாட்சிக்குப் பயப்படுகின்றார்கள்
நேர்மையற்றவர்கள்
சட்டத்திற்குப் பயப்படுகின்றார்கள்.

16. நடைமுறை ‘அன்பு வழி’
ஆகிவிட்டால்,
வன்முறை
முகவரி இழக்கும்

17. நாளைய சுயநலங்களுக்காக
இன்றையப் பொது நலனைப்
புறக்கணித்து வாழ்வில்
என்ன பயன் ஏற்பட முடியும்.

18. விட்டுக்கொடுத்தல்
பலவீனம் அல்ல
விட்டுக் கொடுப்பதால்
ஒருபோதும் தீமை நேராது.

19. ஒரு நாளில்
வருவதல்ல வெற்றி
ஒரு நாள்
வரும் வெற்றி

20. அன்பு செய்வது
கடினம் அல்ல
ஆனால்,
எளிதும் அல்ல!

21. பிறக்கும் உயிர்கள்
அனைத்திற்கும்
உயர்ந்த சிம்மாசனம்
தாயின் மடியே.

22. காலத்தைத் தேடித் தேடி
உழைத்தால்
காலம் உன் பின்னால்
ஓடி வரும்.

23. அடுத்தவர் அரியவில்லை
என்பதனால் மட்டும்
தவறான ஒரு செயல்
நேர்மையாகி விடமுடியாது.

24. இருப்பதை வைத்துக்
கணக்குப் போடுங்கள்
கற்பனையில் உள்ள பணம்
கட்டுமானத்துக்கு உதவாது.

25. சிறந்த தீர்மானத்தை
எடுக்கவில்லையென வருந்தாதே
எடுத்த தீர்மானத்தை
சிறப்பாக நிறைவேற்று.

26. லஞ்சம்...!
ஒரு புற்றுநோய்
நமக்குள்ளே
வளர அனுமதிக்கலாமா?

27. பொய்யான உலகத்தில்
பொய்மையாக வாழ்கின்றோம்
என்ற மெய்மையைப் புரியும் காலமே
மனிதன் மகான் ஆகும் காலம்.

28. காலத்தை நாம்
ஒத்திப் போட்டால்,
காலம் நம்மை
ஒதிக்கிப்போடும்.

29. மனதில் சின்னச் சின்ன வலி
ஏற்படும் போது சோர்ந்து விடாதே!
சில நேரங்களில் சில சங்கடங்களை
சகித்துக் கொள்ளத்தான் வேண்டும்.

30. நடக்கத்
தயங்கினால்
இலக்கை
அடைய முடியாது.

31. எக்காலத்திலும்
ஏமாற்றம் தறாத
நேர்மையான நண்பர்கள்
சிறந்த புத்தகங்கள் தான்.

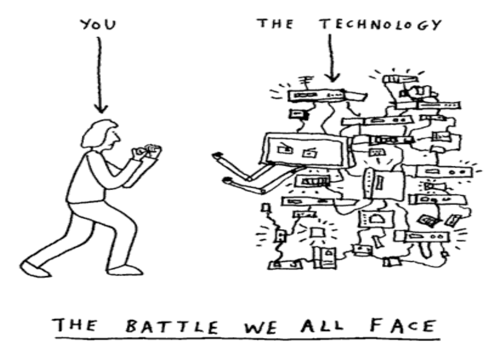
32. கையளவு உள்ளத்தில்
கடலளவு ‘ஆசை’ இருப்பதால்தான்
கடுகளவு பிரச்சனையும்
மலையளவு கனமாகத் தோன்றுகிறது.

33. முடிவு
தெரியாத வரை
முயற்சியை
நிறுத்தக்கூடாது.

34. யார் சொன்னது
நீ தோற்று விட்டாய் என்று
வெற்றி பெறுவதற்கல்லான முயற்சி செய்தாய்
இன்னொரு முறை முயற்சித்தால் போதுமே!

35. உன் சாதனைகளுக்கு
நீ மட்டுமே சொந்தக்காரன்
என்று கர்வம் கொண்டு விடாதே!
சமூகப் பங்களிப்பை மறந்து விடாதே!

36. ஒருவரைப் புரிந்து கொள்ள வேண்டும்
என்று நீங்கள் எண்ணினால்,
முதலில் அவரை
அன்பு செய்யுங்கள்.



37. ஒரு நல்லவனைப் பார்க்க
ஆசைப்படுவதை விட ஒரு நல்லவனாக
இருக்க ஆசைப்படுவோம்.

38. உன்னுடைய இயலாமையே
உனக்கு பலம், ஆணிவேர்
இயலாமையில் இருந்து தான்
நீ படைக்கப் போகிறாய்.

39. பொய் பேசுவதை விட
உண்மையைப் பேசவும்
உண்மையாக வாழவும்
அதிக மனோதிடம் தேவை.

40. அமைதியும் பொறுமையும்
அறிவின் வெளிப்பாடு
கோபமும், பொறாமையும்
உணர்ச்சியின் வெளிப்பாடு.

41. திட்டமிட்ட
உணவுப் பழக்கம்
திட்டமிடாத
நோய்களிலிருந்து பாதுகாக்கும்.

42. உன் நாவில் இருந்து
ஒரு சொல் வெளிப்படும் முன்பே
அதன் விளைவைப் பற்றிய சிந்தனை
உன் மனதில் இருக்க வேண்டும்.

43. மெழுகுவர்த்தியின் உயரத்தில்
மாறுபாடு இருக்கலாம்
வெளிப்படுத்தும் ஒளி ஒன்றே!
பெற்றோரின் தியாகமும் அப்படியே!

44. நமக்குக் கிடைக்கும் குடிநீர்
நம் பேரக் குழந்தைகளுக்கும்
வேண்டுமல்லவா!-நீர் ஆதாரங்களை
ஆக்கிரமிக்காது பாதுகாப்போம்.

திரு. *X.P ஒயிட்*
எழுத்தாளர்

YOUNG INVENTORS

1) **Mission for a vision:**

It pained Jyoti, a STD IX student from DAV Public School, Unit VIII, Bhubaneshwar, to see people with poor eye sight not being allowed to drive. He observed students of a visually impaired school, to see how they managed daily chores. Some of them were suffering from glaucoma.

The idea: He made a CCV- Close circuit Vision gadget which works for glaucoma as well as low vision patients and is has been approved by AIIMS, New Delhi. Now he is planning on making it for paralytic patients.

Future Plans: He aims to become a scientist and specialize in electronics.

2) **No Helmet: No riding**

With road accidents on the rise and people still being very slack when it comes to wearing a helmet, S.M. Arthi, a student of Government Girls H.S.S., Nannilam, and Thiruvarur district in Tamil Nadu decided to it was time to act. A recent accident got them thinking and it was Arthi's visit to a restaurant where she saw the use of a sensor to control the water running from a tap that helped her develop the idea.

The idea: She worked with her friends to use a similar technology to tackle the situation of safety. Using this module when one wants to start the bike he/she will have to wear the helmet. The sensor in the helmet helps the engine start.

Future plans: Arthi wants to become a marine engineer.

Have you ever wondered how you could get people to wear a helmet or help your grandfather climb the stairs without him struggling? And have you done anything about it? The National Innovation Foundation saw over 4104 submissions from across the country to handle problems such as these. IGNITE 11 was organized by NIF in association with Central Board of Secondary Education (CBSE), Society for Research and Initiatives in Sustainable Technologies and Institutions (SRISTI), various State Education Boards and other partners. The best ideas and innovations were awarded by Dr. A. P. J. Abdul Kalam.

3) **Others who won:**

Sajid Ansari from Jharkhand for her desktop machine to Sort rice grain. **Rimjhim Baruah** from Assam for her Poster writing Pen. **Vignesh, Manoj** and **Raghav** from Tamil Nadu for their navigation system to help the visually impaired. **Bhanwar Malviya** from Rajasthan for a drain pipe cleaning device. **Dhavala** from Karnataka for her bed sheet squeezer

Remind me

If you are the type who keeps forgetting to take your medicines on time then Mohit Singh's medicine box reminder will be handy. A Std X student of DPS, Nigahi M.P., decided to design two useful inventions' both of which are practical and necessary.

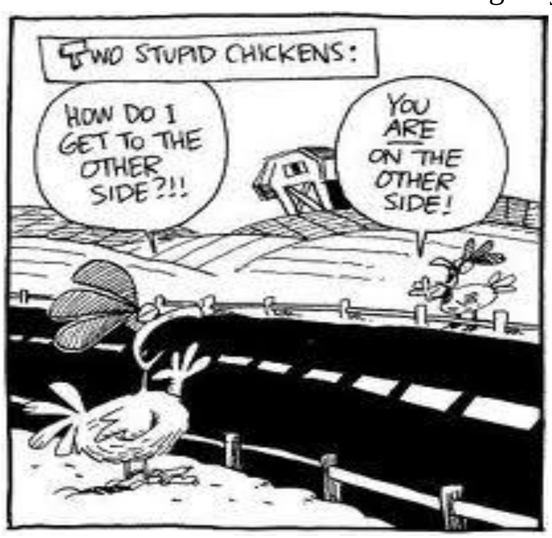
The idea: For taps with different timer settings his idea came from the fact that water was being wasted. He fitted sensors that worked on the principle of time and auto cut off. It's just an accessory which needs to be added to control the use of water. As for the medicine box he set up a LCD panel with a "set time" option which starts ringing when you have to take your medicines. It also has the option of having a BP machine and a sugar level machine.

Future plans: Mohit aspires to become a surgeon.

4) Safe than sorry

Manu Chopra, a Std XI student of G.D. Goenka School, Rohini, Delhi, decided to invent something that will help a victim get away.

The idea: He wanted to create a watch for women. A trendy watch with its dial made up of blue LED lights. Working on the principle of nerve control velocity, this watch can signal the brain to act in case of an emergency.



When the nerve speed increases to 119m/sec as opposed to 60m/sec the dial turns red and the watch gives a shock when it comes in contact with any exposed part of the attacker. A shock of about 0.1 amps is transmitted to allow the person to escape. The watch is also being designed to send a message and a photograph to the three numbers listed.

Future Plans: Manu aspires to be like Steve Jobs. Start his own company.

5) Take it easy

Shalini's grandfather used a walker. She noticed that while he could use the walker on a level surface he found it difficult to walk up the stairs or travel. It is then that this STD IX student of Hartman High School, Patna, decided to make a modified walker with adjustable legs.

The idea: The front legs of the walker are adjustable and can be shortened to climb stairs. With the use of a clip the person walking can adjust the legs.

Future plans: Shalini wants to become a doctor and is still deciding on her specialization.

USE OF GATE SCORES

Some of the Public Sector Units consider the GATE scores for recruitment. They are as follows:

RAILWAY DEPARTMENT

SAIL	Steel Authority of India
BSNL	Bharat Sanchar Nigam Ltd.
BHEL	Bharat Heavy Electricals Ltd.
GAIL	Gas Authority of India Ltd.
ISRO	Indian Space Research Organization
NTPC	National Thermal Power Corporation
NHPC	National Hydroelectric Power Corporation
ECIL	Electronics Corporation
DRDO	Defence and Research Development Organization
DRDL	Defence Research and Development Laboratory
IOCL	Indian Oil Corporation Ltd
EIL	Engineers India Limited
NPCIL	Nuclear Power Corporation of India Limited
NTPS	National Test Pilot School
PGCIL	Power Grid Corporation of India Limited
ONGC	Oil and Natural Gas Corporation
BARC	Bhabha Atomic Research Centre
HAL	Hindustan Aeronautics Ltd.
MTNL	Mahanagar Telephone Nigam Ltd.
BALCO	Bharat Aluminum Company

-Magazine team